

Impact of uncertainties in outflow boundary conditions on the predictions of hemodynamic simulations of ascending thoracic aortic aneurysms

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ABSTRACT

The impact of outflow boundary conditions on the results of numerical simulations of the flow inside ascending thoracic aortic aneurysms is investigated. The adopted outflow conditions are based on the lumped three-element Windkessel model, in which it is assumed that the pressure at the outflow is related to the flow rate and to a terminal constant pressure, through an electric circuit analogue that has a proximal resistance in series with a parallel arrangement of a capacitance and of a distal resistance. The values of the Windkessel model parameters must be a-priori specified. The impact on the numerical simulation results of uncertainties in the values of the Windkessel model parameters is quantified here through a stochastic approach. The propagation of the considered uncertainties is evaluated, in particular, for the instantaneous and time-averaged wall shear stress. The generalized Polynomial Chaos is used as a surrogate model to obtain continuous response surfaces of the quantities of interest in the parameter space starting from a few deterministic simulations. A patient-specific geometry, obtained through *in-vivo* imaging, is considered. The analysis is carried out for both rigid and compliant vessel walls. Our results show that, although the uncertainties in the selected outflow parameters may give significant variability of the instantaneous shear stresses in regions characterized by flow recirculation or large streamline curvature, the impact on cycle-averaged shear stresses is moderate. Taking into account the wall compliance seems to reduce the impact of the uncertainties in the outflow parameters compared to the rigid case.

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1. Introduction

An Ascending Thoracic Aortic Aneurysm (ATAA) is a permanent dilatation of the ascending thoracic aorta and it is conventionally associated to an increase in diameter greater than 50% [1]. ATAAs are life-threatening diseases since they can lead to major complications, like aortic dissection and rupture. The mortality rate is between 15% and 30% in case of Type A aortic dissection [2], i.e., dissection originating in the ascending aorta, and even > 94% if aortic rupture occurs [3]. On the other hand, surgical repair is not safe either, as elective surgery carries a mortality risk of approximately 5% [4]. Therefore, ATAA clinical management should be based on the balance between the risks of the complications caused by the disease and those related to the surgical operation.

Currently, the risk of ATAA complications is evaluated by means of the maximum diameter [5] or its growth rate [6]. Surgery is generally recommended when the ascending aortic diameter reaches 5.5 cm [4,7]. Indeed, Davies et al. [5] found that the yearly risk of serious complications (dissection, rupture or death) dramatically jumps to 14% when the maximum diameter reaches 6 cm. However, Pape et al. in a subsequent paper [8] reported that nearly 60% of the ascending aortic dissections occur with a diameter smaller than 5.5 cm and thus the ATAA size does not seem a reliable predictor, especially in case of small and medium-sized ATAAs.

From a biomechanical point of view, the aneurysmatic pathology has been investigated considering the structural change of the artery wall and its effects on the wall stress distribution. Rupture and dissection can be considered as the effect of preponderant mechanical stresses on the structurally compromised wall and many studies indicated wall stresses as a key quantity for predicting aneurysm evolution (see e.g., [3,9–11]).

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Nevertheless, these structural studies neglect the impact of the hemodynamic flow inside the thoracic aorta since they assume that the only effect of the blood flow is a uniform pressure. Abnormal flow patterns and abnormal values of the wall shear stress, i.e., the stress produced due to the friction between the blood and the arterial wall, may play a key role in the origin and the evolution of the aneurysm [12]. In particular, despite representing a small percentage of the total wall stress, the wall shear stress can have an important effect on endothelial cells, possibly altering their morphology and orientation [13] and their permeability. Indeed, Shang et al. [14] recently found correlation between time-averaged wall shear stress and aneurysm progression, for Type B dissections (i.e., dissection involving the descending region of the aorta).

In this scenario, Computational Fluid Dynamics (CFD) has been used in the last few years to build complex computer models of the cardiovascular system and has enabled the investigation of pressure and flow field at a temporal and spatial resolution unachievable by any clinical methodology. Therefore, CFD allows to easily quantify a variety of variables and indicators difficult to be obtained from *in-vivo* measurements. In addition, since medical images are now capable to provide not only accurate morphological features but also the non-invasive quantification of blood flow [15,16], their combination with CFD simulation techniques permits to investigate hemodynamics on a patient-specific basis. For instance, anatomic data from Computed Tomography (CT) or Magnetic Resonance Imaging (MRI) and/or velocities from phase-contrast MRI have been used to perform patient-specific computational hemodynamics in healthy [17,18] and diseased subjects [19], in order to provide evidence of a possible relation of local wall shear stress with aortic disease [20,21].

Nonetheless, reliable patient-specific models are usually not easy to be obtained, because the accuracy of CFD predictions depends on many different factors. First of all, there are modeling assumptions, such as those concerning the constitutive properties of blood and of vessel walls. Moreover, from a practical viewpoint, the set-up of a patient-specific simulation is characterized by many sources of uncertainty. The main ones concern the domain geometry and boundary conditions. Indeed, the vessel geometry reconstructed from medical images can be affected by different geometrical errors. Moreover, only a portion of the aorta is available and can thus be simulated; hence, inflow and outflow boundary conditions must be prescribed, which are also patient specific and can only seldom be obtained from *in-vivo* measurements, that, when available, are often characterized by a space and time resolution not adequate for numerical simulation. Therefore, the quantification of how these uncertainties in simulation set-up and input parameters propagate into the output quantities of interest, and, in particular, on wall shear stresses, becomes mandatory in order to obtain reliable hemodynamic models.

The problem of propagating uncertainties in cardiovascular applications has been addressed in the literature in hemodynamic studies. Examples are the assessment of the effect of different sources of uncertainties in idealized Abdominal Aortic Aneurysm (AAA) models with rigid walls [22,23], of uncertainties from clinical data in virtual surgery predictions [24], of uncertainties in inlet flow rate on wall shear stresses in healthy thoracic aortas [25] and intracranial aneurysms [26] or of uncertainties in inflow conditions and proximal arterial stiffness for a vascular network [27].

Our work focuses on evaluating the impact of uncertainties in the outflow boundary conditions in a patient-specific ATAA geometry, for both rigid and deformable-wall cases. The choice of outflow boundary conditions is challenging, since they must accurately represent the effect of downstream organs and vessels. Time-varying pressure or velocity measurements are only seldom available simultaneously for all the outflow surfaces; moreover, in some applications, as e.g., treatment planning, the flow rate and pressure at

the outflow surfaces are part of the desired solution. The formulation of proper outflow conditions for hemodynamics has therefore received a significant attention and a few approaches have been proposed in the literature (see e.g., [28] for a detailed overview). An increasingly common practice is to couple 3D hemodynamic simulations to 0D lumped models, which specify time-varying outflow conditions neglecting the spatial variation of parameters and variables. A popular example is the 3-element Windkessel model, in which the outflow pressure is related to the flow rate and to a distal pressure through an electric circuit analogue that has a proximal resistance in series with a parallel arrangement of a capacitance and of a distal resistance. The capacitance mimics the wall compliance, while the proximal and distal resistances represent the viscous resistance the blood flow undergoes inside large and small vessels downstream the considered computational domain. Examples of applications of this approach to hemodynamic simulations can be found in [29–31] and in the references therein. This approach has the advantage to be simpler and less computationally demanding than coupling 3D simulations with 1D models, while being more accurate than classical outflow boundary conditions, such as prescribed pressure or traction-free conditions, which only take into account the resistance to flow in the branches of the computational domain. The values of the elements of the Windkessel model are not a-priori known and they are usually guessed in order to obtain a physiological pressure waveform in the 0D model. However, it is not clear how this choice affects the results of the simulation and a systematic analysis is not present in the literature to our knowledge. A systematic deterministic quantification of the influence of these parameters in patient-specific realistic ATAA simulations is difficult because of the significant cost of each simulation. An efficient alternative is to use a stochastic approach, in which the unknown Windkessel parameters are considered as random variables with a given probability distribution. The uncertainty can, thus, be propagated through the CFD model and a continuous response surface of the output quantities of interest in the parameter space can be recovered through a ‘surrogate’ model, which requires a limited number of deterministic simulations.

In the present work, we use the generalized Polynomial Chaos (gPC) approach [32]. Since a preliminary analysis for the 0D model showed that the proximal resistance and the capacitance are the main parameters affecting the pressure profile, these two quantities were selected as uncertain parameters for the uncertainty quantification analysis, while a fixed value of the distal resistance was maintained. A patient-specific geometry of ATAA acquired by Magnetic Resonance Imaging was considered, and the effect of wall compliance was also investigated. The impact of the uncertainties in the selected outflow parameters was investigated on different quantities of interest. In particular, we analyzed the stochastic mean and standard deviation of the instantaneous and time-averaged wall shear stresses.

2. Methodology and computational set-up

2.1. Modeling and numerical methodology

We consider here blood as a Newtonian and incompressible fluid, as widely accepted for blood in large vessels. Thus, the governing equations take the following form:

$$\begin{aligned} \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} - \nabla \cdot (2\nu \nabla^s \mathbf{v}) + \nabla p &= \mathbf{0} \quad \mathbf{x} \in \Omega, t > 0 \\ \nabla \cdot \mathbf{v} &= 0 \quad \mathbf{x} \in \Omega, t > 0 \end{aligned} \quad (1)$$

where ρ and ν are blood density and kinematic viscosity, respectively, $\mathbf{v}$ is the velocity, p the pressure divided by density, $\nabla^s \mathbf{v}$ the symmetric part of the velocity gradient tensor, $\nabla \mathbf{v}$, and Ω the

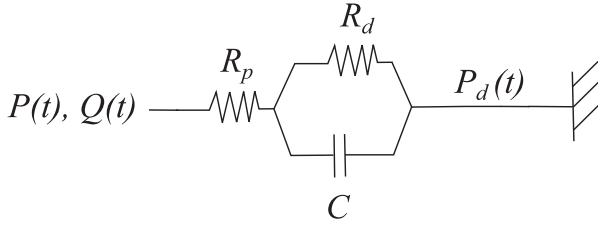


Fig. 1. Scheme of a 3-element Windkessel model.

three-dimensional computational domain. No explicit turbulence modeling has been introduced in the momentum equation.

Blood density and kinematic viscosity can be specified as $\rho = 1.06 \text{ g cm}^{-3}$ and $\nu = 3.77 \times 10^{-2} \text{ cm}^2 \text{ s}^{-1}$.

At the inlet section of the computational domain we specified a Dirichlet boundary condition on the velocity, imposing a flow rate waveform (see e.g., [12,30]).

At the outflows, a time-dependent pressure is imposed, which is calculated through the lumped 3-element Windkessel model, also called RCR model [30] and sketched in Fig. 1. In this model the outflow pressure, $P(t)$, is related to the flow rate, $Q(t)$, and to a distal pressure, $P_d(t)$, through an electric circuit analogue that has a proximal resistance, R_p , in series with a parallel arrangement of a capacitance, C , and of a distal resistance, R_d . The capacitance mimics the wall compliance, while the proximal and distal resistances represent the viscous resistance the blood flow undergoes inside large and small vessels downstream the considered computational domain. This leads to the following differential equation:

$$P(t) + R_d C \frac{dP(t)}{dt} = (R_p + R_d)Q(t) + R_p R_d C \frac{dQ(t)}{dt} + P_d(t) + R_d C \frac{dP_d(t)}{dt} \quad (2)$$

The values of the elements of the Windkessel circuit must be a priori given, as well as the value of the distal pressure. The distal pressure is assumed here to be constant and equal to zero, as often done in the literature. The parameters C , R_d and R_p will be the object of the stochastic sensitivity analysis described in the following.

On the arterial wall we impose a no-slip condition between the fluid and the wall, i.e., $\mathbf{v} = \mathbf{v}_w$. Both rigid ($\mathbf{v}_w = 0$) and deformable ($\mathbf{v}_w \neq 0$) simulations were performed. The equation governing the vessel wall behavior is the following:

$$\rho_w \frac{\partial^2 \mathbf{v}_w}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma}_w + \mathbf{b}_w \quad \mathbf{x} \in \Omega_w, t > 0 \quad (3)$$

where $\mathbf{v}_w$ is the wall displacement field, ρ_w and $\boldsymbol{\sigma}_w$ the vessel wall density and stress tensor, respectively, and $\mathbf{b}_w$ is the volume force acting on the wall surface, Ω_w . The thin wall approximation allows $\mathbf{b}_w$ to be expressed in terms of the wall thickness h_w and of the surface traction $\mathbf{t}_w$ acting on the wall, which is opposite to that ($\mathbf{t}_f$) acting on the fluid: $\mathbf{b}_w = \mathbf{t}_w/h_w = -\mathbf{t}_f/h_w$. In the present work, the vessel wall is modeled as isotropic, homogeneous, elastic and having a constant thickness. The stress tensor for a linear and elastic material can be expressed as $\boldsymbol{\sigma}_w = \mathbf{C} : \boldsymbol{\epsilon}_w$, where $\mathbf{C}$ is the fourth-order stiffness tensor and $\boldsymbol{\epsilon}_w$ is the infinitesimal strain tensor, i.e., $\boldsymbol{\epsilon}_w = \nabla^s \mathbf{v}_w = 1/2(\nabla \mathbf{v}_w + (\nabla \mathbf{v}_w)^T)$. Using the symmetry of $\boldsymbol{\sigma}_w$, the fourth-order stiffness tensor $\mathbf{C}$ can be reduced to a second-order tensor $\mathbf{D}$. For an incompressible, isotropic solid augmented by transverse shear stresses we obtain:

$$\mathbf{D} = \frac{E_w}{(1 - \lambda_w^2)}$$

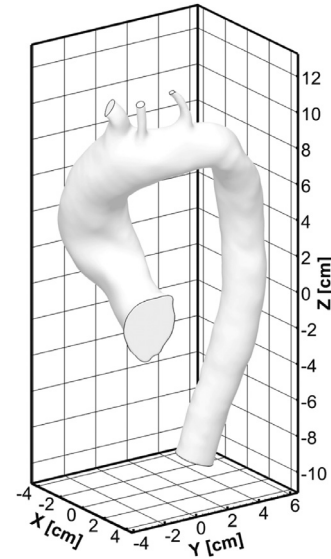


Fig. 2. Sketch of the considered geometry ($D_r = 1.5$).

$$\times \begin{bmatrix} 1 & \lambda_w & 0 & 0 & 0 \\ \lambda_w & 1 & 0 & 0 & 0 \\ 0 & 0 & 0.5(1 - \lambda_w) & 0 & 0 \\ 0 & 0 & 0 & 0.5k_w(1 - \lambda_w) & 0 \\ 0 & 0 & 0 & 0 & 0.5k_w(1 - \lambda_w) \end{bmatrix}$$

where E_w is the Young modulus, λ_w is the Poisson ratio, and k_w is a parameter accounting for a parabolic variation of the transverse shear stress through the membrane [33].

The hemodynamic simulations were carried out with *SimVascular*, an open-source comprehensive package specific for cardiovascular problems (see [34] for reference). Eq. (1) is discretized by means of a stabilized finite element method. SUPG/PSPG stabilizing terms are added since the standard Galerkin formulation is subjected to instabilities due to the convective term and to the mixed character of the problem [35]. The stabilized formulation makes it possible to choose linear shape functions for both velocity and pressure (so-called P1-P1 elements). Additional stabilizing terms are then introduced to prevent also the numerical instabilities due to backflow divergence at the outlets, which is typical of cardiovascular applications [36].

When taking into account the wall compliance, the solver enforces a linearized kinematics approach, in which the fluid mesh is kept fixed but the nodes at the interface with the solid wall have in general non-zero velocities. The strong form of the mechanical problem for the vessel wall is given in Eq. (3). The solid and the fluid problems are coupled through the thin wall approximation previously described. Therefore, the conventional finite element formulation is modified accordingly. The effect of the vessel wall results in a Neumann-type boundary condition for the fluid at the interface, thus leading to a single variational form for the coupled problem [33].

For advancing in time, the generalized α -method is used, which is an implicit technique that allows to achieve second order accuracy in time (see [37] for further information).

2.2. Simulation set-up

A patient-specific geometry of thoracic aorta subjected to aneurysm in its ascending region was considered. The patient object of the study is male with tricuspid aortic valve, age 65 years, weight 74 kg. The geometry is sketched in Fig. 2 and was acquired by MRI. The maximum diameter ratio $D_r = D_{\max}/D_{\text{healthy}}$, where

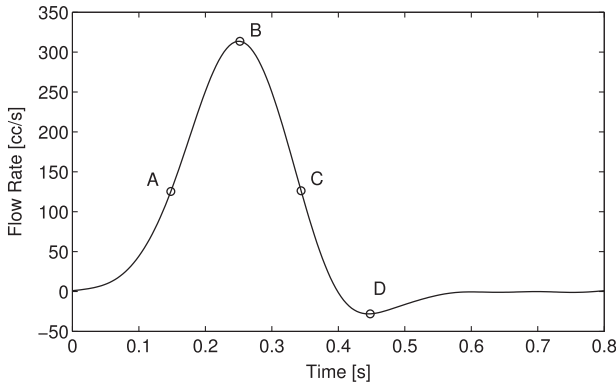


Fig. 3. Flow rate profile imposed at the inlet section. The following characteristic time instants are highlighted: the maximum acceleration (A), the peak systole (B), the maximum deceleration (C) and early diastole (D).

$D_{\max}$ is the diameter measured in correspondence of the widest section of the aneurysm and D_{healthy} a diameter characteristic of the healthy segment, is $D_r = 1.5$. The maximum diameter is $D_{\max} = 3.8$ cm.

As for inflow boundary conditions, we imposed the physiological profile shown in Fig. 3 and assumed a uniform distribution of the velocity in the inlet section. As a further assumption, only the velocity component normal to the inlet section was considered, as often made in this type of simulations (see e.g., [17,38]). Input patient-specific data could be used to prescribe the flow rate waveform, when available from *in-vivo* measurements. In the present study, since the focus is on the effect of the outflow conditions, we used the same ‘guessed’ physiological profile of Fig. 3 for all the simulations performed. The resulting Reynolds number, based on the diameter of the inlet section and on the bulk inlet velocity averaged during the cardiac cycle, was $Re \approx 700$.

Boundary conditions based on the three-element Windkessel model, described in Section 2.1, were imposed at each outlet. The values of the parameters contained in the model are usually set in order to match patient-specific clinical data [39]. Such information was not available for the present work, therefore as a first guess we derived the parameters to obtain a desired physiological behavior of the pressure profile (125/80 mmHg) in the simplified fully lumped model in which the contribution of the computational domain is not taken into account [30]. More specifically, Eq. (2) was solved numerically until the combination of R_p , C and R_d produced the desired behavior of $P(t)$. The resulting values are: $R_p = 106.5$ g cm⁻⁴ s⁻¹, $C = 10.4 \times 10^{-4}$ cm⁴ s² g⁻¹ and $R_d = 1.67 \times 10^3$ g cm⁻⁴ s⁻¹.

Since descending from a fully lumped parameter model, the so obtained values of R_p , R_d and C are independent of the particular geometry considered, only depending on the flow rate waveform imposed at the inlet. They were then distributed in the various outlets as follows [38]:

$$\begin{aligned} R_{p_i} &= R_p \frac{A_{\text{tot}}}{A_i} \\ R_{d_i} &= R_d \frac{A_{\text{tot}}}{A_i} \quad \text{for } i = 1, \dots, n_{\text{outlets}} \\ C_i &= C \frac{A_i}{A_{\text{tot}}} \end{aligned} \quad (4)$$

where n_{outlets} is the number of the outlets (i.e., 4 in the considered case), A_i is the area of the outlet i and A_{tot} is the sum of the areas of all the outlets. The previous relations arise from the consideration that when increasing the vessel section the resistance opposed to the blood flow becomes smaller, whereas the wall compliance capability increases. Unlike R_p , R_d and C , the quantities obtained through Eq. (4) are geometry-dependent.

Table 1

Summary of mechanical properties of the arterial wall employed in deformable simulations.

Quantity	Value
Density: ρ_w [g cm ⁻³]	1.08
Young's Modulus: E_w [MPa]	3
Thickness: h_w [cm]	0.1
Poisson's Ratio: λ_w	0.499
Shear Correction Parameter: k_w	0.833

As previously said, when considering the arterial compliance, the aorta was assumed to be a linear, elastic, homogeneous and isotropic material, with a uniform wall thickness. The values of the mechanical properties chosen are reported in Table 1, where all dimensional properties, except for Young's Modulus, are expressed for convenience in CGS units.

An unstructured computational grid consisting of tetrahedral elements was used. A grid sensitivity study was carried out by considering four grids, having approximately 1.1×10^5 , 1.6×10^5 , 2.4×10^5 and 3.9×10^5 nodes. The details about the procedure used to check grid independence is presented in Appendix A. Based on this sensitivity analysis, the grid with 2.4×10^5 nodes, 3.3×10^4 of which on the aortic wall, was chosen. This grid consists of 1.4×10^6 almost isotropic elements, 6.6×10^4 of which on the wall. The number of the elements at inlet and at the descending aorta outlet section is 1.5×10^3 and 6.3×10^2 , respectively. The number of elements in the outlet sections of the three small branches is approximately 60.

The physical time step is 0.001 s for both rigid and deformable cases, resulting in 800 time steps every cardiac cycle.

In order to make sure that the flow reached periodicity after an initial transient, for each computation we evaluated the L_2 norms of the normalized differences between two successive pressure waveforms and between two successive flow rate waveforms. These quantities were evaluated in correspondence of the descending aortic outlet section, and the simulations were run until the above L_2 norms were smaller than 10^{-3} [40]. We usually needed to simulate ~ 10 cardiac cycles in order to satisfy this criterion.

Finally, the discretization of the governing Eq. (1) with the previously described finite-element method, which implies the introduction of a numerical viscosity, might be considered as an implicit large-eddy simulation, in which the scales of turbulence larger than the grid size are simulated, while the numerical viscosity acts as a subgrid scale model (see e.g., [41]). In order to check the possible presence of resolved turbulent fluctuations, we analyzed the time signals of the flow variables in specific locations, characterized by significant curvature of streamlines and flow acceleration (see Appendix B for the details of the analysis). For all the considered grid resolutions we did not notice significant deviations from a periodic behavior, meaning that no turbulence fluctuations corresponding to turbulence scales resolvable on the considered grids were found.

2.3. Uncertainty quantification methodology

We briefly recall here the main features of generalized Polynomial chaos, which is the strategy used in the present work to obtain a continuous response surface in the parameter space starting from a few deterministic numerical simulations. The gPC approach, in its non-intrusive form, is based on the projection of a given random process over an orthogonal known basis [32]. The gPC expansion for a stochastic response output, X , may be expressed as follows (term-based indexing):

$$X(\omega) = \sum_{k=0}^{\infty} a_k \Phi_k(\xi(\omega)) \quad (5)$$

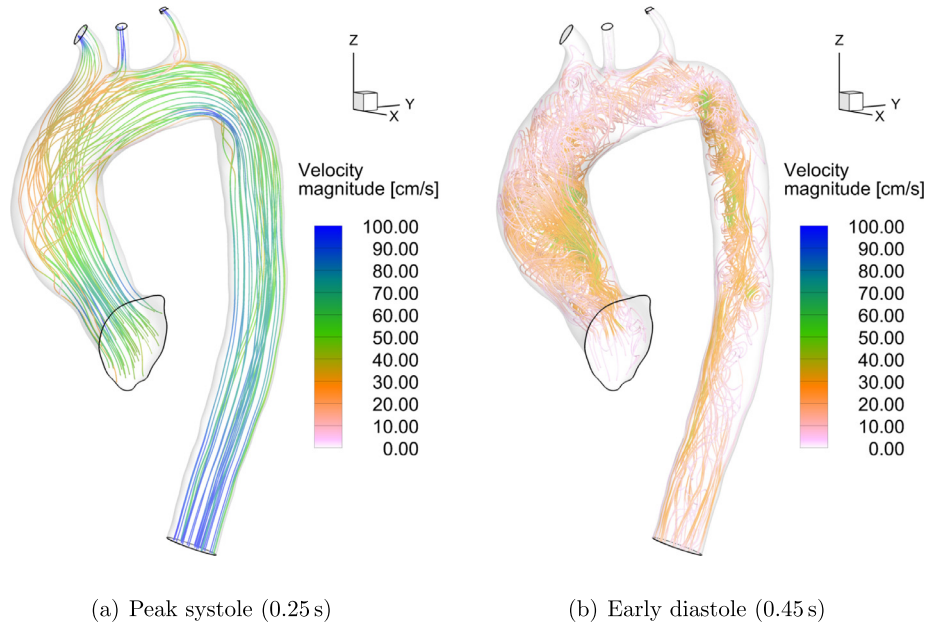


Fig. 4. Instantaneous flow streamlines in case of rigid walls at time instants B and D, as defined in Fig. 3. The streamlines are colored according to the velocity magnitude.

where ω is a random event, $\xi(\omega)$ is the vector consisting of the independent random variables in the event space, Ξ , (corresponding thus to the set of considered uncertain parameters), $\Phi_k(\xi)$ is the gPC polynomial of index k and a_k is the corresponding Galerkin projection coefficient.

The response surface is obtained by a truncation of the expansion (5) to a finite limit T . Applying polynomial order bounds for all one-dimensional polynomials (tensor-product expansion), T is obtained as follows:

$$T = \prod_{i=1}^M (P_i + 1) - 1 \quad (6)$$

where M is the dimension of the random parameter space (i.e., the number of the uncertain parameters) and P_i is the polynomial order bound for the i th parameter.

Thanks to the orthogonality of the polynomial basis, the coefficient a_k can be computed as follows:

$$a_k = \frac{\langle X, \Phi_k \rangle}{\langle \Phi_k, \Phi_k \rangle} = \frac{1}{\langle \Phi_k, \Phi_k \rangle} \int_{\omega \in \Xi} X \Phi_k(\xi) \rho(\xi) d\xi \quad (7)$$

where $\rho(\xi)$ is the weight function associated with the polynomial family chosen. The integrals can be computed through different mathematical methods: Gaussian quadrature is used in the present work. The polynomial family, Φ_k , must be a-priori specified and its choice affects the speed of the convergence of the gPC expansion: unsuitable polynomials may lead to the need of a large number of degrees of freedom (i.e., a higher order of the polynomials) to obtain a given level of accuracy, whereas a suitable polynomial family is able to interpolate both the input and the random variables by means of fewer degrees of freedom. When dealing with Gaussian quadrature, an optimal family has a weight function similar to the probability measure of the random variables. The choice of the polynomial family thus depends on the Probability Density Function (PDF) shape of the uncertain parameters.

As previously explained, we wish to investigate the sensitivity to the parameters contained in the Windkessel model used to impose outflow boundary conditions, i.e., the proximal resistance R_p , the capacitance C , and the distal resistance R_d . Even in the reduced-order gPC approach, the stochastic sensitivity analysis to three uncertain parameters remains extremely expensive, because

each deterministic simulation is 3D and unsteady. Therefore, we carried out a preliminary analysis to evaluate the effects of R_p , C and R_d on the output pressure cycle in the 0D Windkessel model, i.e., after neglecting the contribution of the 3D geometry of the aorta. This analysis (presented in Section 4) allowed to single out the sensitivity to each uncertain parameter and, thus, to select the most important parameters for the Uncertainty Quantification (UQ) analysis on 3D simulations, whose results are discussed in Sections 5.1 and 5.2.

The three uncertain parameters were allowed to vary in the following ranges: $R_p \in [85.8, 148.4] \text{ g cm}^{-4} \text{ s}^{-1}$, $C \in [5.82 \times 10^{-4}, 12.6 \times 10^{-4}] \text{ cm}^4 \text{ s}^2 \text{ g}^{-1}$ and $R_d \in [1.57 \times 10^3, 1.86 \times 10^3] \text{ g cm}^{-4} \text{ s}^{-1}$. These ranges (based on the values obtained in Section 2.2 for the reference deterministic simulations) were chosen such that each uncertain parameter approximately produced the same maximum variation of $P(t)$ in the pressure waveform: we imposed a variation of about 12 mmHg, which represents almost 10% of the desired systolic pressure in the deterministic simulations. This uncertainty is reasonable in our opinion, considering that data on the patient pressure are usually available.

The PDF of the uncertain input parameters was assumed to be uniform. The choice of a uniform PDF is motivated by the fact that, among the classically used distributions, it is the least informative distribution with the highest variance in given intervals. Therefore, it is usually adopted when there is lack of information on the input parameter distribution functions, as in our case and in most cases when dealing with simulation parameters. It is expected that, for a given variation interval of the input parameters, the uniform PDF distribution should give the largest variability of the output quantities, thus providing a conservative estimation of the sensitivity to the considered input parameters. As a consequence of the assumed uniform input PDF, Legendre polynomials were used for the gPC basis. The polynomial expansion was truncated to the third order for each dimension; hence, four quadrature points for each random variable were sufficient to compute the coefficients of the expansion. These values are shown for all the parameters in Table 2. An analysis of the convergence of the truncated gPC expansion is reported in Appendix C.

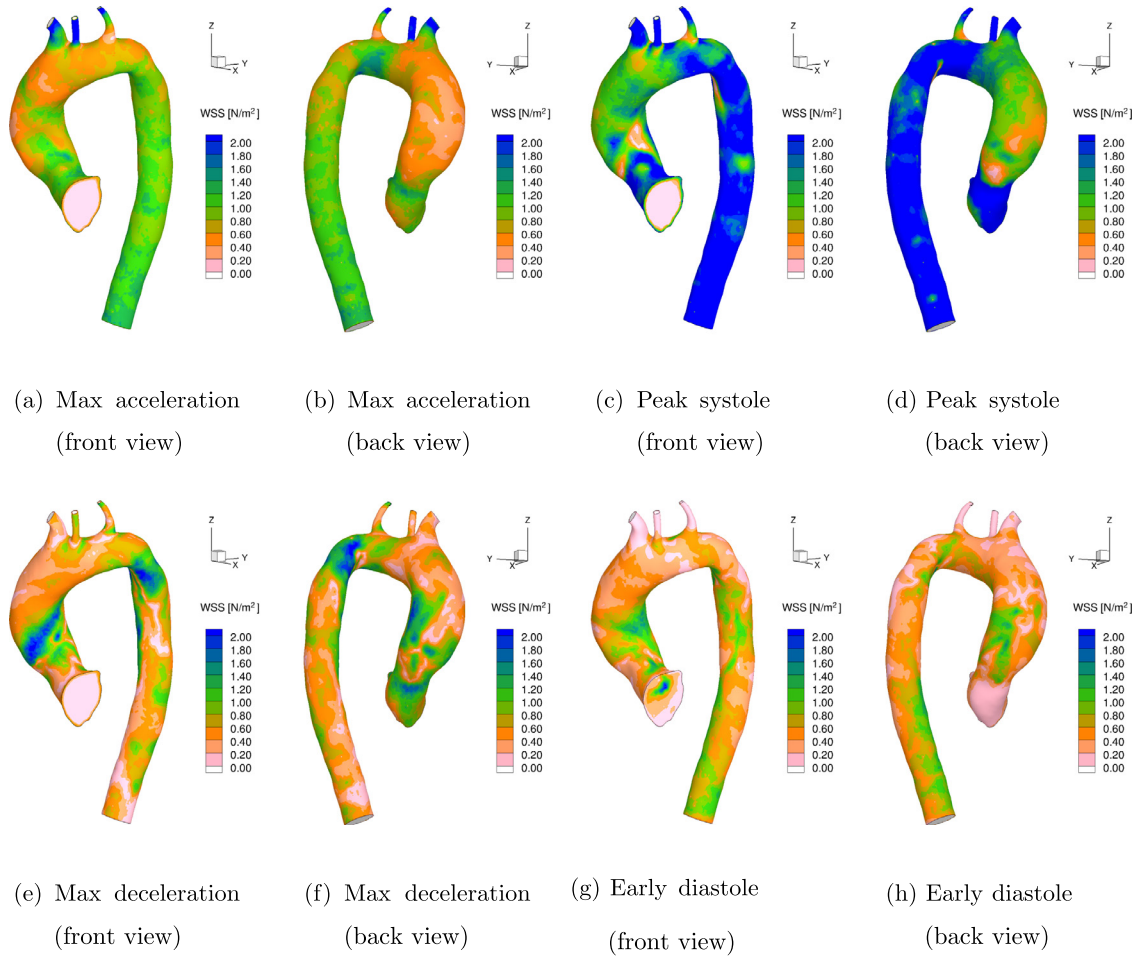


Fig. 5. WSS distribution in the deterministic simulation with rigid walls.

Table 2

Quadrature points for each input parameter (R_p , C , R_d).

Random variable	1st	2nd	3rd	4th
R_p [$\text{g cm}^{-4} \text{s}^{-1}$]	90.1	106.5	127.7	144.1
C [$\text{cm}^4 \text{s}^2 \text{g}^{-1}$]	6.29×10^{-4}	8.05×10^{-4}	10.4×10^{-4}	12.1×10^{-4}
R_d [$\text{g cm}^{-4} \text{s}^{-1}$]	1.59×10^3	1.67×10^3	1.76×10^3	1.84×10^3

3. Results of deterministic simulations

This section briefly describes the main flow features observed in the deterministic simulations carried out with the reference values of the outflow RCR parameters given in Section 2.2. Both rigid and deformable walls were considered.

When considering the effects of the blood flow on the arterial wall, particular attention is given to the Wall Shear Stress (WSS), namely the modulus of the tangential stress τ exerted by the flow on the arterial wall, i.e.,

$$\text{WSS}(\mathbf{x}, t) = |\boldsymbol{\tau}(\mathbf{x}, t)| \quad (8)$$

since this quantity is thought to play an important role in the formation and growth of vascular diseases, as e.g., aneurysms [42,43]. The Time-Averaged Wall Shear Stress (TAWSS), i.e., the mean wall shear stress averaged throughout an entire cardiac cycle of period T , is often used as an hemodynamic descriptor [28]. It is defined as follows:

$$\text{TAWSS}(\mathbf{x}) = \frac{1}{T} \int_0^T |\boldsymbol{\tau}(\mathbf{x}, t)| dt \quad (9)$$

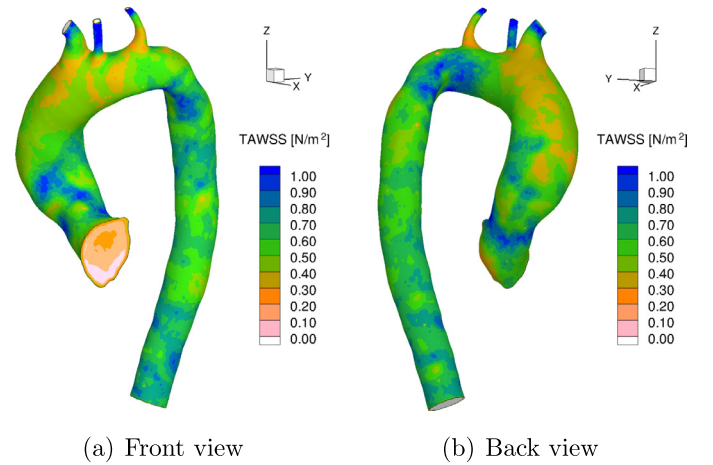


Fig. 6. TAWSS distribution in the deterministic simulation with rigid walls.

3.1. Rigid walls

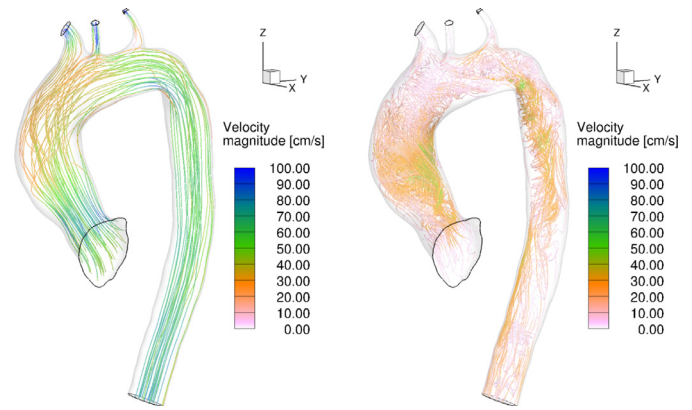
Fig. 4 shows the instantaneous flow streamlines obtained in the simulation of the flow with the assumption of rigid walls. The streamlines are colored according to the velocity magnitude. During systole they show a very regular behavior and the highest values of the velocity (Fig. 4(a)), while Fig. 4(b) shows that the instantaneous streamlines largely deviate from the mainstream flow direction in the diastolic phase, which is also characterized by local



Fig. 7. Three main regions of the aorta: ascending aorta, aortic arch and descending aorta (the branches are not considered).

recirculations and by reverse flow entering from the domain outlets. In the ascending region, in which the aneurysm is located, a noticeable local deviation of the streamlines from the mainstream direction can be observed also during peak systole. This is most probably due to the geometry of the aorta, which requires the flow to make a large deviation for reaching the aortic arch region from the inlet section. In our simulations we did not introduce any flow swirl at the inlet section, since only the normal component was considered.

In Fig. 5 the WSS distributions at different time instants are shown. The four considered time instants are those indicated with markers in Fig. 3. As expected the WSS at the systolic peak and in the acceleration phase of the inlet flow cycle are generally significantly larger than those characterizing the time instants corresponding to the deceleration phase. Note how largest wall shear stresses in the diastolic phase are found in the internal parts of the aneurysm and of the distal aortic neck (see Fig. 5(g) and (h)).

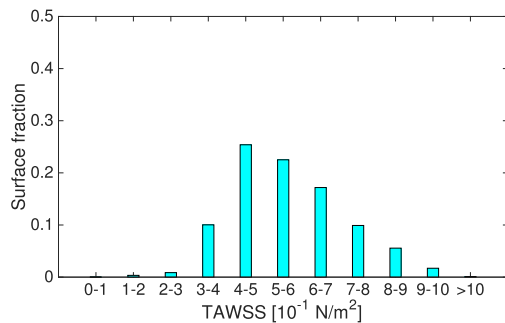


(a) Peak Systole (0.25 s)

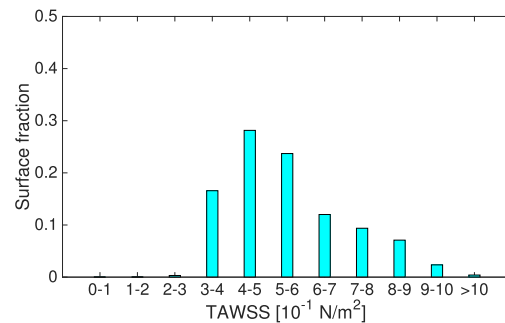
(b) Early Diastole (0.45 s)

Fig. 9. Instantaneous flow streamlines in case of deformable walls at time instants B and D, as defined in Fig. 3. The streamlines are colored according to the velocity magnitude.

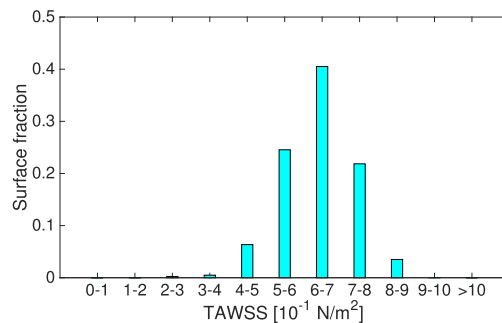
TAWSS is presented in Fig. 6. The TAWSS distribution gives information on the mean behavior of the WSS during an entire cardiac cycle, and it is an average representation of what we found in the instantaneous maps: the most stressed regions in average are the small branches and regions characterized by significant curvature of the vessel geometry. In order to better quantify the stresses and their distribution, we divided the geometry in three main regions: the ascending aorta, the aortic arch and the descending aorta (see Fig. 7). The small branches are not considered in the present analysis. We then identified 11 stress levels, from level 1 ($0\text{Pa} \leq \text{TAWSS} < 0.1\text{Pa}$) to level 11 ($\text{TAWSS} \geq 1\text{Pa}$), all the intermediate levels having a range of 0.1 Pa. For each region, the surface fractions associated to the various stress levels are presented in Fig. 8, which allows a more quantitative appraisal of the TAWSS distribution shown in Fig. 6. For instance, it can be seen that the



(a) Ascending aorta



(b) Aortic arch



(c) Descending aorta

Fig. 8. Surface fraction of the TAWSS levels in the deterministic simulation with rigid walls.

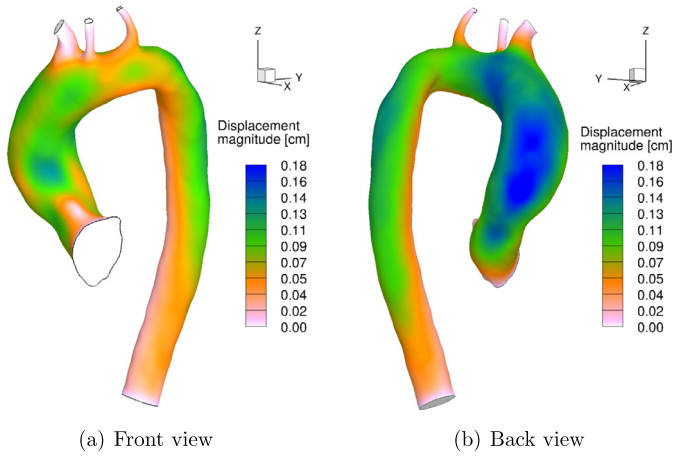


Fig. 10. Displacement field at peak systole; the arterial wall was modeled as an elastic, homogeneous and isotropic material, with uniform wall thickness.

descending aorta has a larger surface fraction subjected to high stress levels (Fig. 8(c)). In addition, in the descending region there are fewer significant stress levels, which indicate a more uniform distribution of TAWSS. Conversely, both the aortic arch and the ascending aorta, including the aneurysm region, are characterized by a large variability of TAWSS, which, however, on the largest part of the surface takes lower values than in the descending aorta.

3.2. Deformable walls

In this section we evaluate the effect of the compliance of the arterial wall, modeled as a linear, elastic, homogeneous and isotropic material, with uniform wall thickness (see Section 2.2).

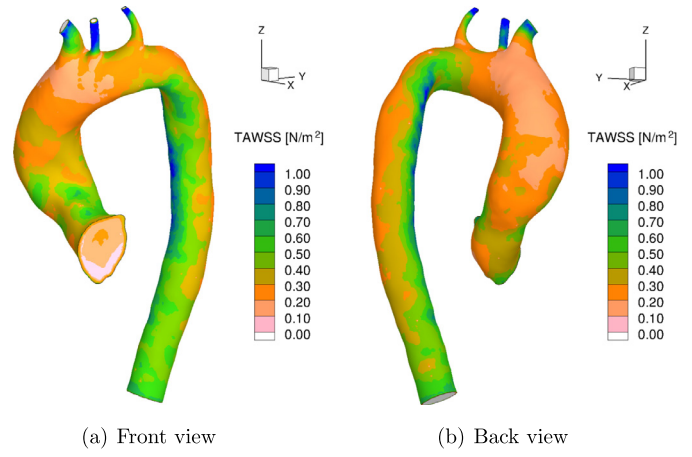


Fig. 11. TAWSS distribution in the deterministic simulation with deformable walls.

The flow streamlines do not present significant differences if compared with those obtained with rigid walls. A slight decrease of the velocity magnitude is observed, especially in the descending region (compare Fig. 9(a) with Fig. 4(a)). This behavior is due to the widening of the vessel under the action of the blood flow. Fig. 10 shows the displacement field at peak systole, which is the time instant when the deformations are most significant. The maximum displacements are found in the region of the aneurysm, where they reach about 5% of the maximum aortic diameter. The values of the displacement are referred to the vessel nominal geometry.

Fig. 11 shows the TAWSS distribution: the deformable-wall simulation presents lower stress values in most part of the aorta, in particular in the distal aortic neck and in large part of the aneurysm and of the descending region. However, the internal part

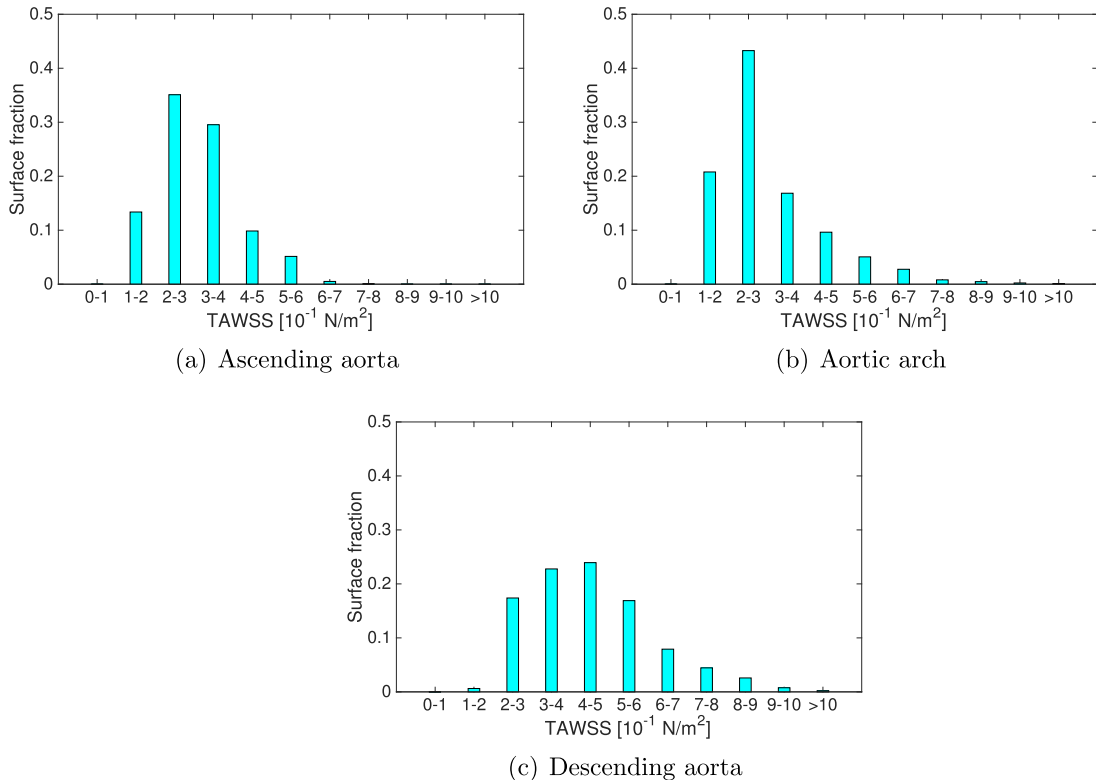


Fig. 12. Surface fraction of the TAWSS levels in the deterministic simulation with deformable walls.

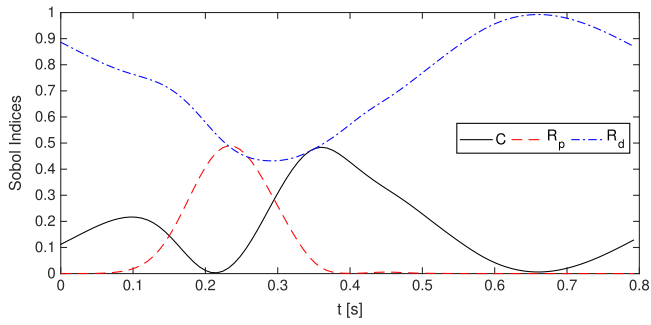


Fig. 13. Partial sensitivities to the RCR parameters of the pressure waveform obtained in the OD model.

of the descending aorta has larger values of TAWSS in the deformable case (compare Fig. 11(a) with Fig. 6(a)). Fig. 12 shows the distribution of the TAWSS levels in the three regions introduced in Section 3.1, which confirm the previous observations. All regions are in general characterized by lower stresses if compared with the rigid case (compare Figs. 12 and 8), while, the comparison between Figs. 12(c) and 8(c) indicates that, consistently with the previous observations and with Figs. 11 and 6, in the descending aorta the deformable case presents significantly larger variability of TAWSS.

4. Preliminary UQ analysis of the effect of uncertainties in the RCR parameters on the pressure cycle of the OD Windkessel model

In this section we summarize the results of the preliminary stochastic study about the effects of the RCR parameters on the output pressure cycle in the OD Windkessel model, i.e., without

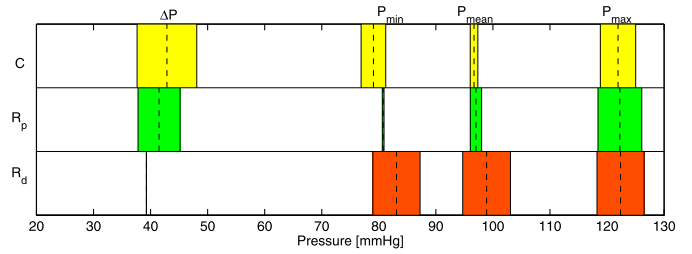


Fig. 15. Stochastic mean (dashed line) $\pm$ stochastic standard deviation (continuous lines) of the systolic pressure $P_{\max}$, of the diastolic pressure $P_{\min}$, of the mean pressure P_{mean} , and of the pulse pressure ΔP , obtained from the OD model by varying a single parameter of the RCR model at a time.

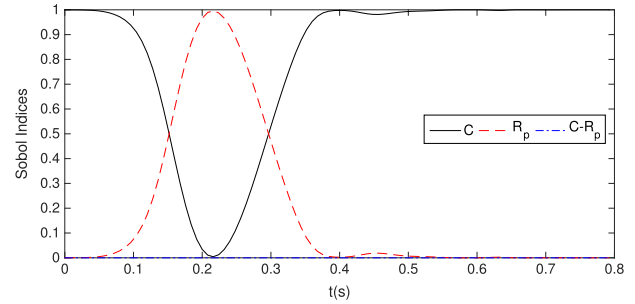


Fig. 16. Partial sensitivities to the C and R_p of the pressure waveform obtained in the OD model.

considering the actual flow dynamics inside the considered aorta segment. The three parameters were considered simultaneously and allowed to vary in the ranges introduced in Section 2.3, thus resulting in 64 evaluations of the OD model. Fig. 13 reports the pressure partial sensitivities to the RCR parameters and indicates

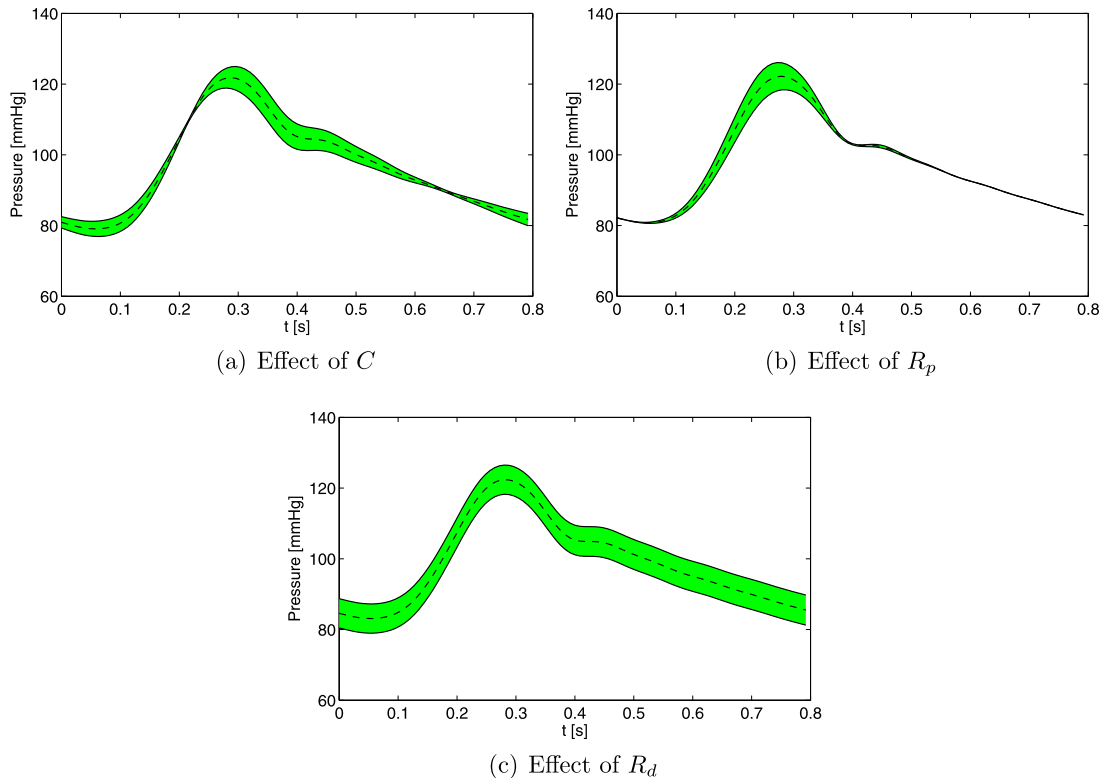


Fig. 14. Stochastic mean (dashed line) $\pm$ stochastic standard deviation (continuous lines) obtained from the OD model by varying a single parameter of the RCR model at a time.

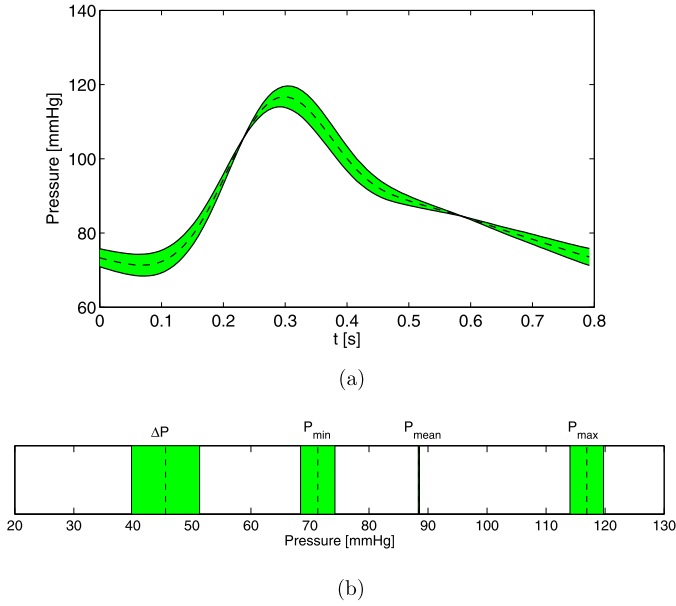


Fig. 17. Effect of the variation of the capacitance C on: (a) the pressure waveform in correspondence of the descending aorta outlet section and (b) the systolic pressure $P_{\max}$, the diastolic pressure $P_{\min}$, the mean pressure P_{mean} , and the pulse pressure ΔP . Rigid vessel walls. The results are expressed in terms of stochastic mean (dashed line) $\pm$ stochastic standard deviation (continuous lines).

a prevalent effect of the distal resistance R_d in most part of the cardiac cycle. The interaction effects are not reported in the figure since they were found to be negligible. This suggests that the sensitivity of the system to the unknown parameters can be derived

from an analysis in which the variation of the RCR parameters is considered individually.

Fig. 14 shows the pressure waveform in terms of the stochastic mean value $\pm$ the stochastic standard deviation obtained by varying a single parameter at a time. In addition, Fig. 15 highlights the sensitivity of specific quantities of the pressure profile, viz. the systolic pressure $P_{\max}$, the diastolic pressure $P_{\min}$, the mean pressure P_{mean} and the pulse pressure ΔP , again to the variation of a single parameter at a time.

It can be seen that the capacitance C has a large impact on the pulse pressure: when the value of C increases then $P_{\max}$ decreases and $P_{\min}$ increases, thus producing a smaller pulse pressure. On the other hand, the mean pressure value P_{mean} is practically constant (see Fig. 15). The proximal resistance mainly changes the maximum pressure, which increases together with the value of R_p (see Figs. 14(b) and 15). Finally, the distal resistance R_d has an almost constant effect all over the cardiac cycle, which is consistent with the high partial sensitivity to this parameter found in Fig. 13. However, R_d produces a simple translation of the pressure waveform: when increasing the value of R_d also the values of $P_{\max}$, $P_{\min}$ and P_{mean} increase, whereas the pulse pressure ΔP does not change (see Figs. 14(c) and 15).

Since only R_p and C are responsible for changing the shape of the pressure waveform, they were selected as the uncertain parameters for the following UQ analysis. In order to make a preliminary evaluation of their relative effect, Fig. 16 shows the sensitivity of the 0D pressure waveform to these two parameters (R_d maintained fixed). The parametric coupling is again negligible, whereas both parameters have a significant impact in some regions of the cardiac cycle, with a prevalence of the capacitance C .

Therefore, in the UQ analysis we first evaluated the effect of C alone (Section 5.1), for both rigid and deformable cases. Then,

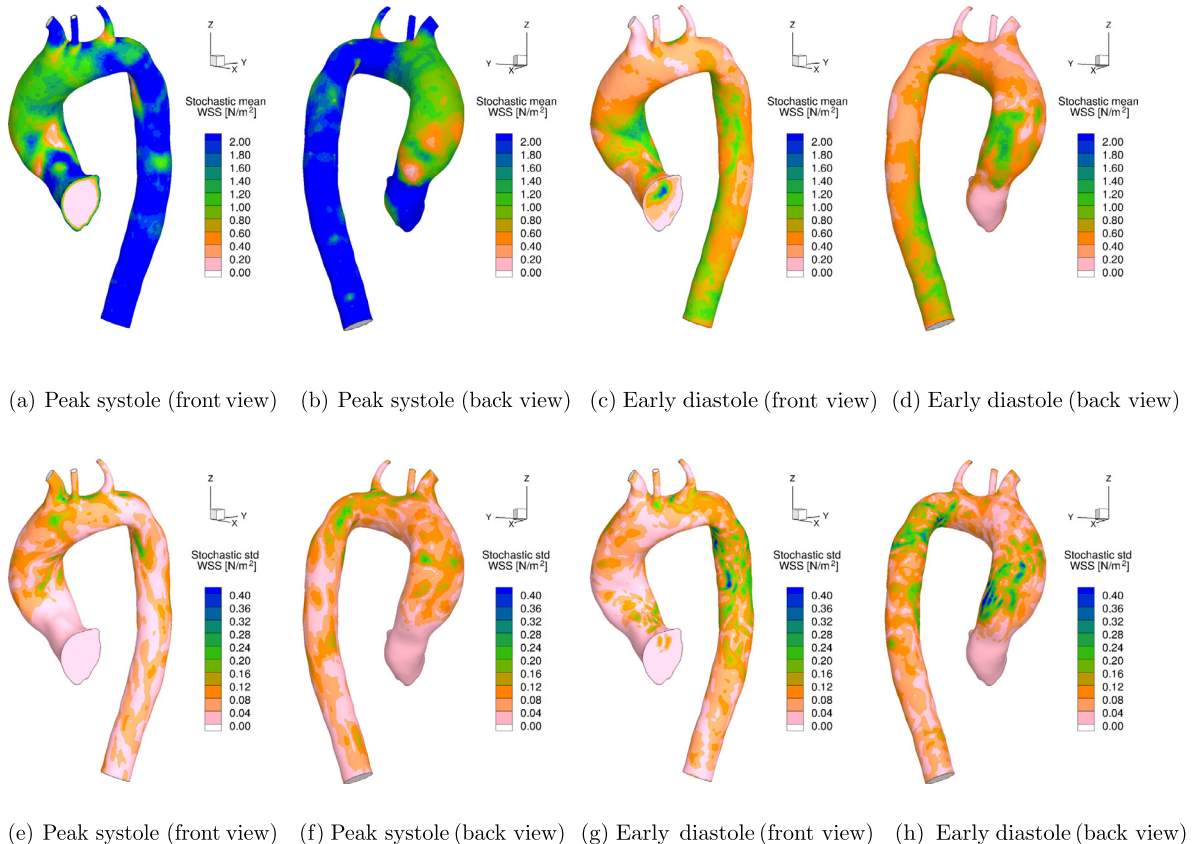


Fig. 18. Stochastic mean value (a–d) and stochastic standard deviation (e–h) of WSS in case of rigid walls. Effect of uncertainties in the value of the capacitance C .

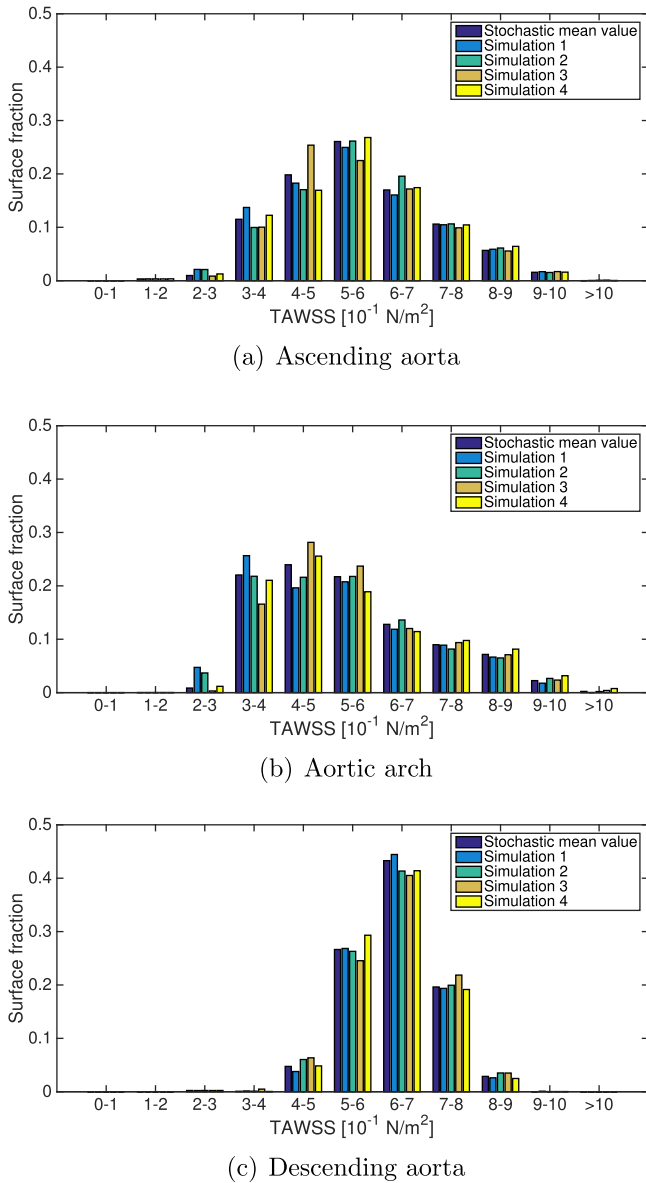


Fig. 19. Surface fraction of the TAWSS stochastic mean value levels compared with that obtained for each deterministic simulation (Simulation 3 is the reference simulation). Effect of uncertainties in the value of the capacitance C in case of rigid walls.

we investigated the combined effect of C and R_p (Section 5.2); in order to limit the computational costs, in this case only the rigid case was considered.

During all these analyses, a reference value was maintained for the distal resistance: as in Section 2.2, we chose $R_d = 1.67 \times 10^3 \text{ g cm}^{-4} \text{ s}^{-1}$.

5. Sensitivity to the outflow parameters of 3D hemodynamic simulation results

5.1. Effects of uncertainties in the value of C

In this Section we evaluate how an uncertainty in the value of the capacitance C affects the quantities of interest. C was allowed to vary in the range given in Table 2, gPC expansion truncated to the third order is used to build the response surface in the param-

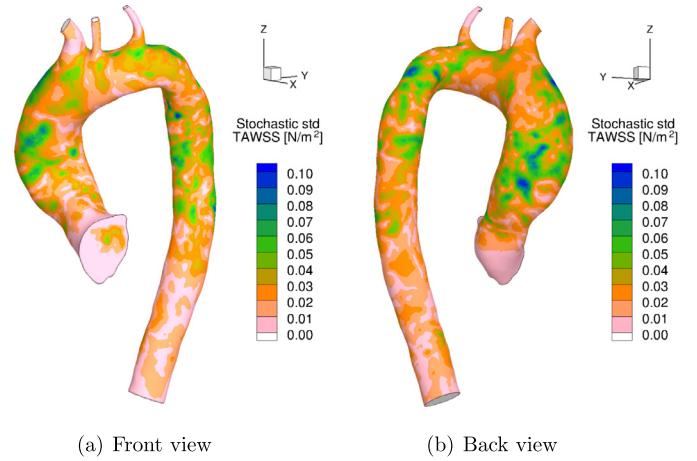


Fig. 20. Stochastic standard deviation of TAWSS (a–b) in case of rigid walls. Effect of uncertainties in the value of the capacitance C .

eter space and this results in 4 deterministic hemodynamic simulations for each case considered.

Section 5.1.1 provides the results obtained in case of rigid walls, Section 5.1.2 discusses the effect of the wall compliance.

5.1.1. Rigid walls

Fig. 17(a) shows the output pressure waveform obtained in correspondence of the descending aorta outlet section. The results are presented in terms of the stochastic mean value $\pm$ the stochastic standard deviation. This profile can be compared with the one previously found in the OD sensitivity analysis (see Fig. 14(a)). The two waveforms are similar, except for a slight reduction of the values of the pressure curve when the effect of the aorta is taken into account. Thus, as found in Section 4 an increase of the capacitance C reduces the pulse pressure, as systolic and diastolic pressures become closer.

More quantitatively, the same effect of C as in Section 4 can be observed also on the characteristic quantities of the pressure cycle, viz. $P_{\max}$, $P_{\min}$, P_{mean} and the pulse pressure ΔP (compare Fig. 17(b) with the upper panel of Fig. 15). The stochastic mean values of $P_{\max}$, $P_{\min}$ and P_{mean} are slightly different if compared with those found with the OD model, all reduced by approximately 7 mmHg. Consequently, the stochastic mean value of the pulse pressure ΔP is almost the same. The stochastic standard deviations of all the quantities considered do not change significantly compared to the OD case. Therefore, it may be concluded that a OD analysis can give useful indications on the sensitivity of the output pressure waveform.

More interesting is the effect of the variation of C on the hemodynamic output quantities introduced in Section 3. This is evaluated in terms of the mean value and of the standard deviation of the output quantities in the parameter space, also called in the following stochastic mean and stochastic standard deviation.

The stochastic mean value of WSS for the case of rigid walls is shown in Fig. 18(a) and (b), at peak systole and early diastole. As previously said, the stochastic mean value gives the average of the response surface of the considered quantity in the parameter space. These fields are qualitatively very similar to those obtained for the reference deterministic simulation, which corresponds to a single given value of the uncertain parameter C (compare Fig. 18(a) and (b) with Fig. 5(c) and (d) for peak systole and Fig. 18(c) and (d) with Fig. 5(g) and (h) for early diastole). On the other hand, the stochastic standard deviation of WSS, shown in Fig. 18(e)–(h), quantifies the variability of WSS with C . Significant variance is found in both the considered characteristic time instants, but the

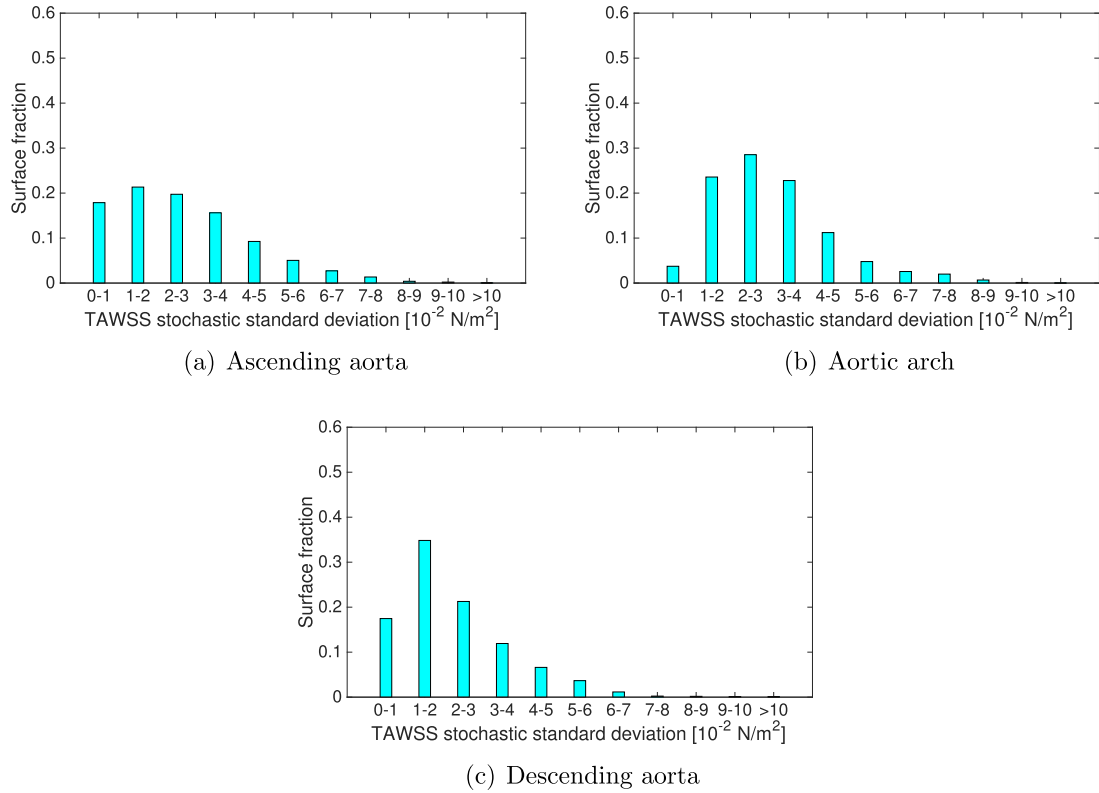


Fig. 21. Surface fraction of the TAWSS stochastic standard deviation levels. Effect of uncertainties in the value of the capacitance C in case of rigid walls.

highest values of the variance, of the order of 0.5 Pa, are observed at early diastole in correspondence of the distal aortic neck and of the ascending aneurysm, which are thus the regions most sensitive to uncertainties in C . These are the regions characterized at early diastole by extended recirculation areas (see e.g., Fig. 4(b) for the reference deterministic simulation). Therefore, it seems that variations in the C outflow parameter may have a significant impact on instantaneous WSS when flow recirculations are present.

As for cycle-averaged stresses, Fig. 19 shows the surface fraction at which the TAWSS values are inside a given interval for each deterministic simulation compared to the stochastic mean value of TAWSS. The same levels shown in Fig. 8 are considered. It can be seen that there is a certain variability of the stress distribution between the different simulations, especially in the aortic arch and in the ascending aorta, and that the stochastic mean value can be close to one or another deterministic simulation. The variability of TAWSS with C is better quantified by the stochastic standard deviation, which is visualized through isocontours in Fig. 20(a) and (b). The variation of TAWSS with C is more limited than that observed for instantaneous stresses, the standard deviation of TAWSS being almost everywhere lower than 0.1 Pa. The areas characterized by the largest standard deviation, and thus by the most significant sensitivity to C , are located in the aneurysm and in the distal aortic neck, where also large variability of instantaneous WSS was observed at the early diastole. As previously discussed, they are connected with flow recirculation regions. As done for the mean stochastic value, we identified 11 levels of standard deviation, the first 10 having a range of 0.01 Pa from 0 Pa to 0.1 Pa, the last level considering TAWSS values larger than 0.1 Pa and we plotted in Fig. 21 the surface fraction associated with each level for the three regions identified in Fig. 7. It can be seen that in all regions, on most of the surface the variability of TAWSS is lower than 0.04 Pa; consistently with the previous observations, the surface fraction characterized by a variability higher than 0.04 Pa is larger for

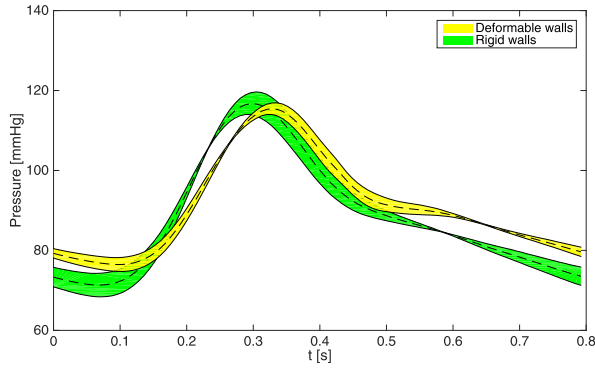
the ascending aorta and the aortic arch than for the descending aorta region.

5.1.2. Deformable walls

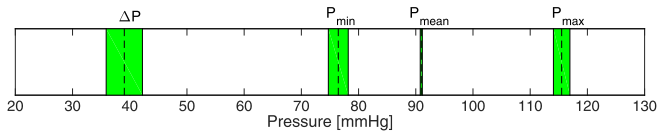
In this section, we investigate the sensitivity to C in simulations with deformable vessel walls. The vessel wall mechanical properties are the same as those used in the deterministic simulation described in Section 3.2.

As before, the pressure waveform is evaluated in correspondence of the outlet section of the descending aorta, and the results are presented in terms of the stochastic mean value $\pm$ the stochastic standard deviation in Fig. 22(a). The results obtained by considering the walls rigid are also reported in the same figure for comparison. It is clear that the main effect of taking into account wall deformations is a reduction of the pulse pressure: this result could be expected, since the wall compliance acts as a further capacitance contribution and in the OD model an increase of C was observed to reduce the pulse pressure, whereas the mean pressure value does not significantly change. This is confirmed by analyzing the single characteristic quantities of the pressure waveform, reported in Fig. 22(b): by comparison with the analogous results obtained for rigid walls and shown in Fig. 17(b) we can notice that the stochastic mean value of ΔP reduces of almost 6 mmHg, while a smaller difference is found in the stochastic mean value of P_{mean} (the change is about 2 mmHg). Another interesting difference is the reduction of the stochastic variance in the compliant case. The wall compliance thus seems to have a damping effect, reducing the variability in the results due to uncertainties in C .

Let us analyze now the sensitivity to C of instantaneous and time-averaged shear stresses. Fig. 23 shows the stochastic standard deviation at peak systole and early diastole. As observed in the simulations for rigid walls, the largest variability is observed at early diastole in regions of the aneurysm and of the aortic arch in which flow recirculation occurs. However, the variability obtained



(a) Pressure waveform, comparison with the case of rigid vessel walls



(b) Systolic pressure $P_{\max}$, diastolic pressure $P_{\min}$, mean pressure P_{mean} , and pulse pressure ΔP

Fig. 22. Effect of the variation of the capacitance C on the pressure at the descending aorta outlet section, deformable vessel walls. The results are expressed in terms of stochastic mean (dashed line) $\pm$ stochastic standard deviation (continuous lines).

in the deformable case is in general significantly lower than that found in the case of rigid walls (recall Fig. 18(e)–(h)).

A smaller variability than in the rigid case is also found for TAWSS. This can be seen, for instance, in Fig. 24(c) and (d), in which the difference between the stochastic standard deviations of TAWSS in the deformable and the rigid cases is reported. Except for small regions, the difference is negative indicating that, as previously said, when the deformations of the wall are taken into account the sensitivity to variations of C is reduced compared to the rigid case. This is further confirmed by Fig. 25, in which the surface fractions corresponding to different levels of the TAWSS stochastic standard deviation obtained with deformable walls is compared to the same quantities obtained in the rigid case. In the deformable case all regions have larger surface fractions subjected to the two lowest stress levels. In particular, in the aortic arch approximately 75% of the surface has values of the stochastic standard deviation smaller than 0.02 Pa, whereas in the rigid case more than 70% of the surface presents values higher than 0.02 Pa (Fig. 25(b)).

5.2. Effects of uncertainties in the values of R_p and C

The aim of this Section is to assess the effect of uncertainties in both the proximal resistance R_p and in the capacitance C on the quantities of interest. R_p and C are allowed to vary in the ranges indicated in Table 2. As previously, gPC truncated at third order in each variable is used to obtain the response surface, thus resulting in 16 hemodynamic deterministic simulations. This analysis only concerns rigid simulations because taking into account moving walls implies much larger computational costs than assuming them rigid, making, thus, the stochastic analysis with two uncertain parameters rather difficult from a practical viewpoint. Moreover, it was previously observed that wall compliance reduced the sensitivity to variations of C . Therefore, the present analysis carried

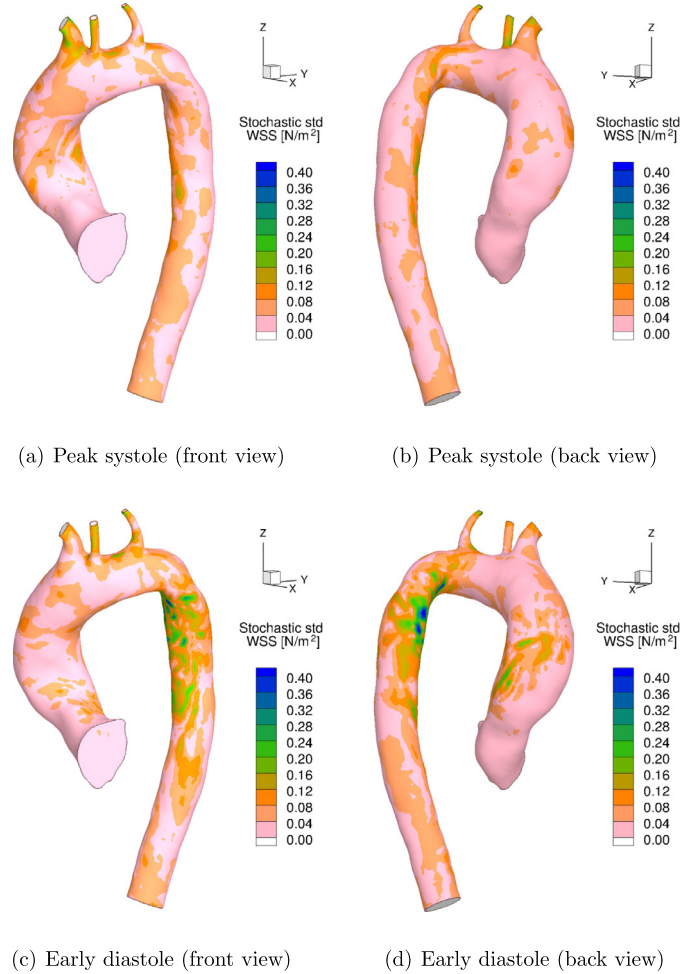


Fig. 23. Stochastic standard deviation of WSS in case of deformable walls. Effect of uncertainties in the value of the capacitance C .

out for rigid walls may be expected to give a ‘conservative’ estimation of the sensitivity to R_p and C .

Fig. 26(a) shows the output pressure waveform at the descending aorta outlet section, again in terms of the stochastic mean value $\pm$ the stochastic standard deviation. The main difference compared to the case in which only the capacitance C was allowed to vary is an increased variability especially in the region near the maximum pressure value (see Fig. 17 for comparison). Fig. 26(b) confirms this indication, presenting the partial sensitivities to the two uncertain parameters alone and combined. For most part of the cardiac cycle the capacitance C is the main responsible of the variability of the results and the uncertainty in the proximal resistance R_p does not produce significant effects on the pressure profile, which thus practically maintains the same behavior as when only C is varied. The sensitivity to R_p becomes predominant only in a small portion of the cycle, near peak systole and this explains why an increased stochastic standard deviation can be observed in the region near the maximum pressure value. It is worth noting that Fig. 26(b) is very similar to Fig. 16, which was obtained in the OD case. The increase in the interaction effect around $t = 0.6$ s is in fact simply due to a negligible total variability, as can be seen from the comparison with Fig. 26(a).

Fig. 27 analyzes the single characteristic pressure quantities, also illustrating the partial sensitivities. It is clear that C is the sole responsible of the diastolic pressure $P_{\min}$, while R_p the sole responsible of P_{mean} . This can be explained recalling that C cannot produce significant changes in P_{mean} and R_p is not capable to vary

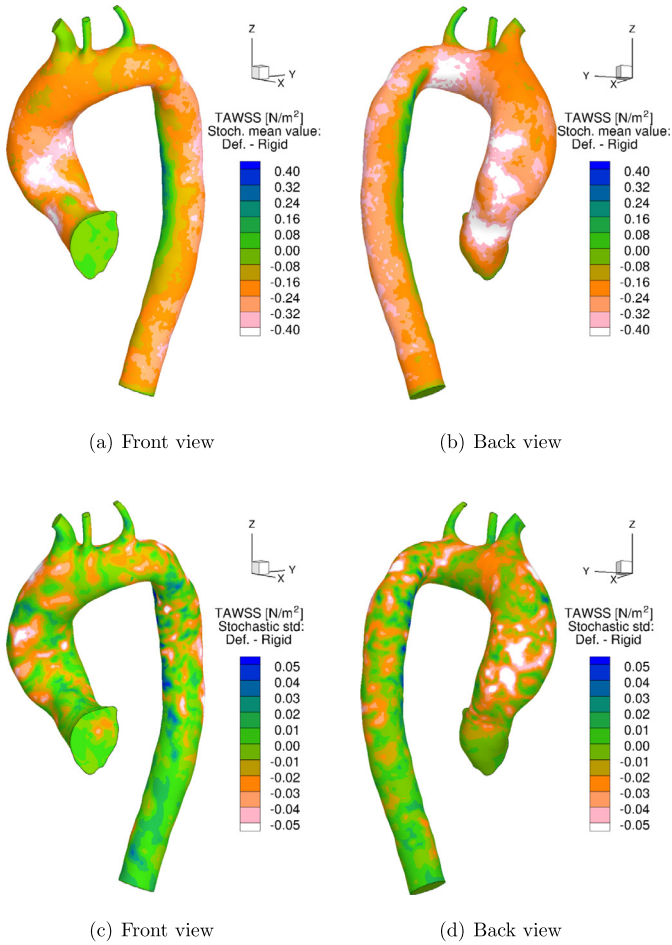


Fig. 24. Effect on the TAWSS distribution of uncertainties in the value of the capacitance C : difference between the stochastic mean values in deformable and rigid cases (a–b), and difference between the stochastic standard deviations in deformable and rigid cases (c–d).

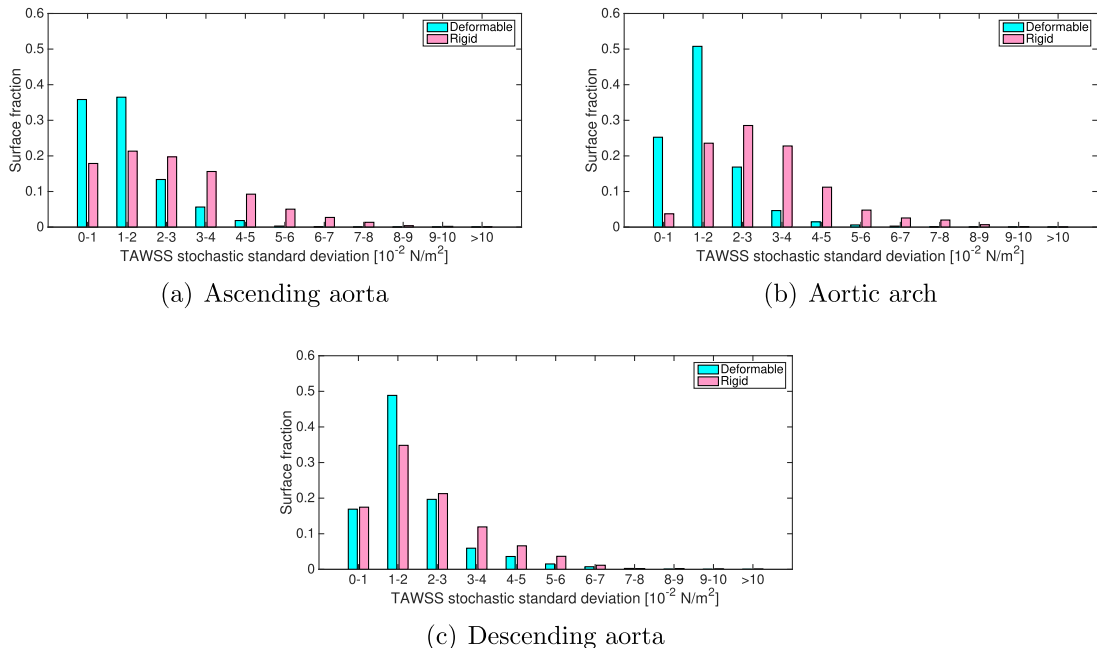


Fig. 25. Surface fraction of the TAWSS stochastic standard deviation level: comparison between deformable and rigid walls. Effect of uncertainties in the value of the capacitance C .

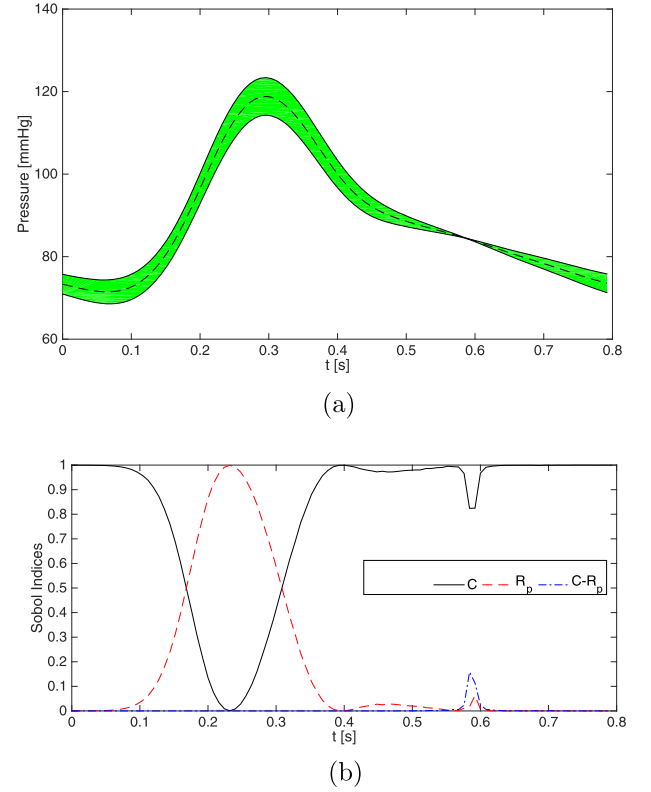


Fig. 26. Effect of the variation of the capacitance C and proximal resistance R_p on the pressure waveform in correspondence of the descending aorta outlet section (rigid walls). The results are expressed in terms of stochastic mean $\pm$ stochastic standard deviation (a) and partial sensitivities to the two parameters (b).

$P_{\min}$ (see Section 4). Both parameters instead affect the other two quantities, with prevalence of C for the pulse pressure ΔP and R_p being the main responsible for systolic pressure $P_{\max}$. The quantities that are affected by R_p have now, for obvious reasons, a larger stochastic standard deviation compared to the case in which only

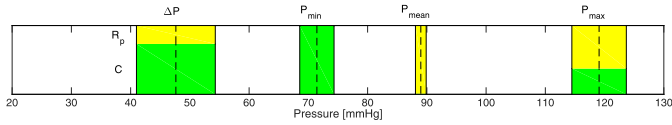


Fig. 27. Effect of the variation of the capacitance C and proximal resistance R_p on the systolic pressure $P_{\max}$, on the diastolic pressure $P_{\min}$, on the mean pressure P_{mean} , and on the pulse pressure ΔP (rigid walls). The results are expressed in terms of stochastic mean $\pm$ stochastic standard deviation. The colors indicate the partial sensitivities to the two parameters: green is the sensitivity to C and yellow to R_p . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

uncertainties in C were considered (Fig. 17(b)). These results are once again consistent with those of the 0D analysis carried out in Section 4.

In order to quantify the variability introduced by the second source of uncertainty on wall shear stresses, Fig. 28(a)–(d) show the isocontours of the difference between the stochastic standard deviation of WSS obtained by varying both C and R_p and that due to uncertainties in C only. As can be expected, the introduction of uncertainties in R_p produces a larger variability at both peak systole (Fig. 28(a) and (b)) and early diastole (Fig. 28(c) and (d)). The increase of variability involves more extended regions at early diastole, especially involving the distal aortic neck, where the difference of stochastic standard deviation reaches even 60% of the value found with uncertainties only in C . As for C , the largest sensitivity to R_p is found in regions which are characterized by significant streamline curvature and flow recirculation, as e.g., the aortic neck at early diastole.

The averaged effect during a cardiac cycle that R_p produces on WSS is illustrated by the difference between the stochastic standard deviation of TAWSS obtained by varying both C and R_p and that corresponding to uncertainties only in C (Fig. 28(e) and (f)). The differences are lower than for instantaneous stresses, but the largest values are located approximately in the same regions. Fig. 29 shows the surface fractions associated to the levels of TAWSS stochastic standard deviation (defined in Section 5.1). By analyzing Fig. 29(a)–(c) we find that with two sources of uncertainty larger areas present a stochastic standard deviation greater than 0.02 Pa: approximately 80% of the ascending region, 90% of the aortic arch and 60% of the descending aorta (in the previous UQ analysis with a single source of uncertainty these values were: 60%, 70% and 45%, respectively). This confirms that the introduction of uncertainties in R_p leads, as expected, to a largest variability also of TAWSS; however, most of the surface in all the considered regions is characterized by a variability which remains lower than 0.05 Pa and, thus, rather small especially if compared with other sources of uncertainty in numerical simulations.

6. Concluding remarks and discussion

A stochastic analysis has been carried out of the sensitivity of the results of hemodynamic numerical simulations in a patient-specific geometry of ascending thoracic aorta with aneurysm to the parameters of the Windkessel model used to prescribe the outflow conditions. A continuous response surface of the quantities of interest in the parameter space was built starting from a few deterministic simulations by means of gPC. Based on a previous 0D analysis,

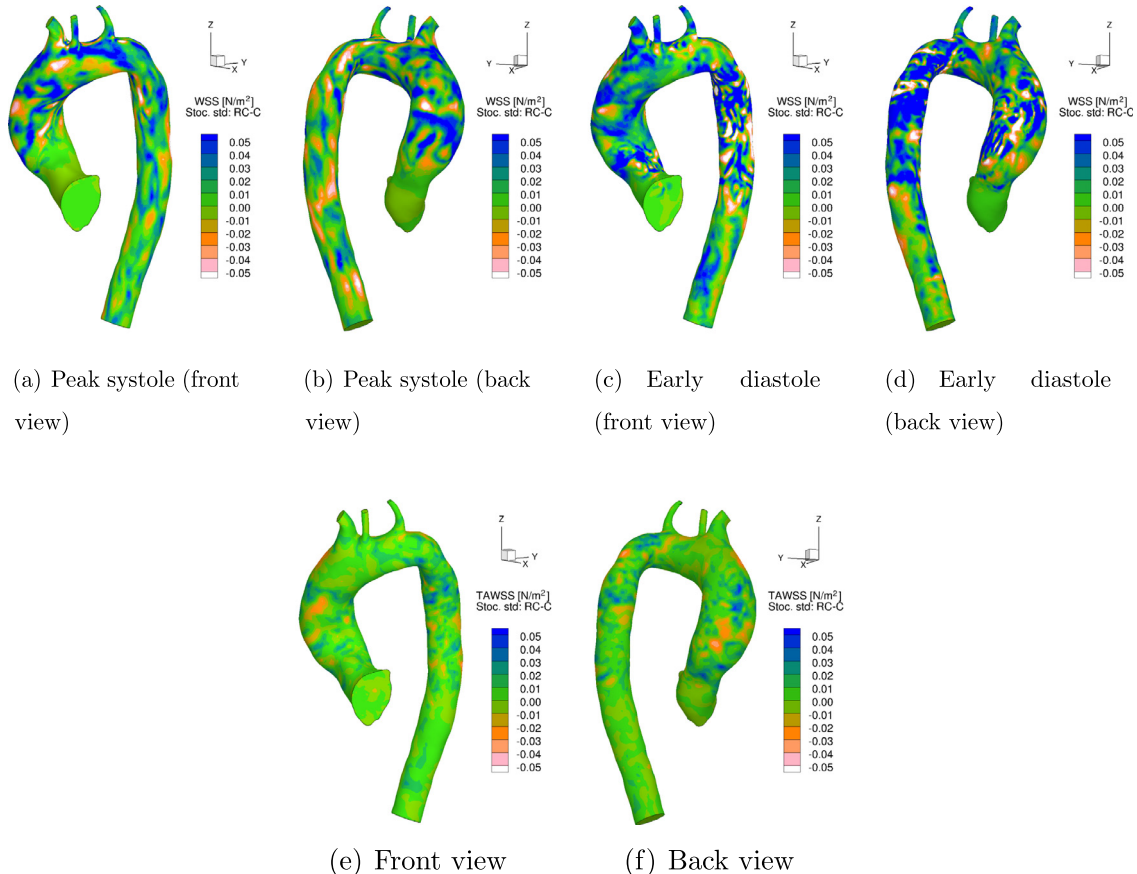


Fig. 28. Difference between the stochastic standard deviations of WSS (a–d) and of TAWSS (e–f) obtained when considering both the uncertainties in C and R_p , and the uncertainty in C alone (rigid walls).

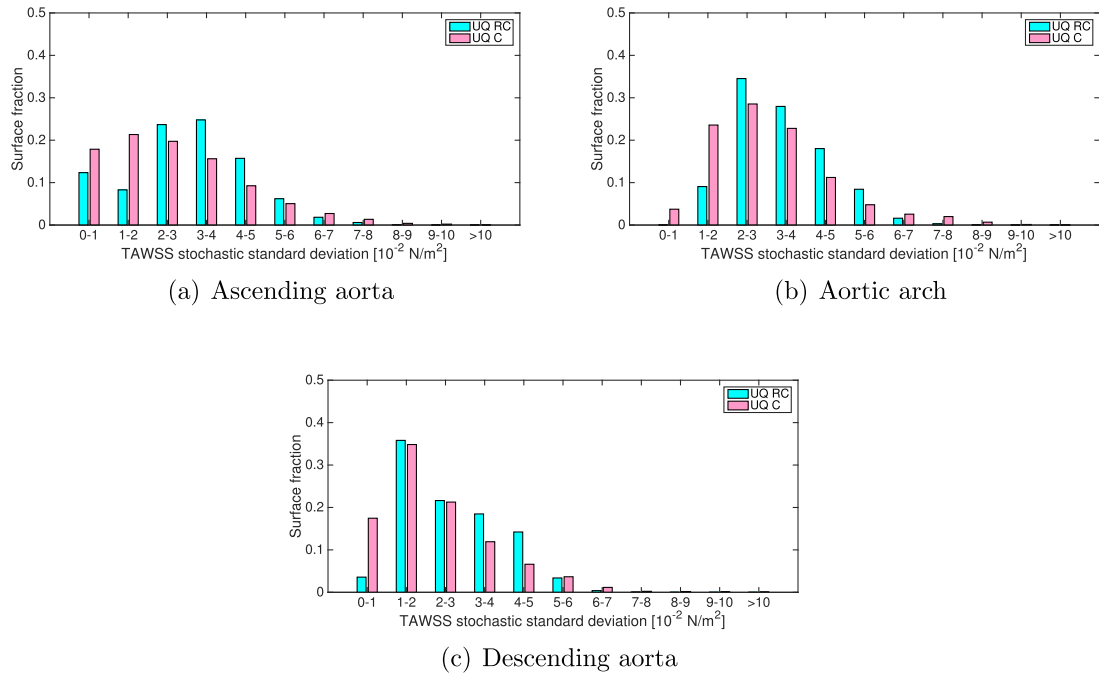


Fig. 29. Surface fraction of the TAWSS stochastic standard deviation levels. Comparison between the UQ analysis with both the uncertainties in the capacitance C and in the proximal resistance R_p , and that with only uncertainties in C (rigid walls).

the capacitance C and the proximal resistance R_p were chosen as uncertain parameters.

Consistently with the OD analysis, the capacitance C has the largest effect on the pressure waveform at the aorta outlet, and in particular on the pulse pressure, while R_p mainly changes the maximum pressure.

More interesting is the effect of uncertainties in these two quantities on wall shear stresses. Significant sensitivity to these uncertainties is observed for instantaneous WSS in regions characterized by large streamlines curvature and flow recirculation, as e.g., in some parts of the aortic neck and of the aneurysm at early diastole. This might have some impact if numerical simulations are used to derive predictions on the aneurysm growth rate, since instantaneous ‘abnormal’ wall-shear stress behaviors of this kind may have significant effects on the vessel tissue properties and, thus, on the aneurysm evolution, as suggested in the literature. Nonetheless, if the WSS average on the entire cycle is considered, the variability remains in all cases low on most of the vessel surfaces. Wall compliance has the effect of further reducing the sensitivity of both mean and instantaneous shear stresses compared to the rigid-wall case.

Similar results were obtained also for different patient-specific geometries, not reported here for the sake of brevity (see [44]). Therefore, the present analysis indicates that the values of the Windkessel parameters seem not to be a major source of uncertainty in the numerical computation of wall shear stresses and, especially, on their average over the cardiac cycle. Therefore, the usual procedure of estimating these parameters in order to obtain a physiological wave pressure form in a OD model seems to be adequate for a correct prediction of wall shear stresses, clearly within all the other sources of error possibly present in the simulations.

The presented results were obtained under some modeling and set-up assumptions. Among the most significant is the choice of the variation ranges and PDF distributions of the uncertain parameters. Patient-specific values of the RCR parameters are hard to be extrapolated from clinical data and they were not available for this study. Therefore, we assumed to have data on the patient pressure,

which are usually available, at least for what concerns the maximum and minimum pressure values during the cardiac cycle, and we chose ranges of variation of the Windkessel parameters leading to reasonable changes in the pressure waveform. If larger variation intervals are assumed, obviously, the variability of the output quantities is expected to become larger. In this context the assumption of a uniform PDF for the uncertain parameters seems to be a reasonable choice because, among the classically used distributions, it is the least informative distribution with the highest variance in given intervals. Therefore, for given variation intervals of the uncertain parameters, the uniform PDF is expected to give the largest variability of the output quantities.

Another issue is the modeling of the arterial wall behavior, which was assumed herein either rigid or linear elastic. The aorta is known to have a non-linear mechanical behavior and more complex models were explored in the literature (see e.g., [19] or [45]). In addition, the aneurysmal wall can have a stiffer mechanical behavior than the ‘healthy’ aortic wall [46]. However, the use of more sophisticated models would lead to computational costs not affordable for the present stochastic study, in which several three-dimensional hemodynamic simulations were needed. Despite the limitations of the used linear elastic model, we expect that the observed trends, and in particular the fact that the sensitivity to the Windkessel parameters is reduced compared to the rigid case, also hold qualitatively for more complex mechanical models. On the other hand, the above discussed refined models are particularly recommended whether the quantitative values of stress and strain are the main objective of the study, e.g., for accurately characterizing the region of the aneurysm.

Finally, other sources of uncertainty are clearly present and some of them may have a significantly larger impact than those analyzed herein, as e.g., uncertainties in the flow rate waveform imposed at the inlet section [25].

Nonetheless, we presented a general methodology which permits to quantify the propagation of the uncertainties in input data on the output quantities of interest in patient-specific ATAA configurations. In the present case, we focused on uncertainties in the

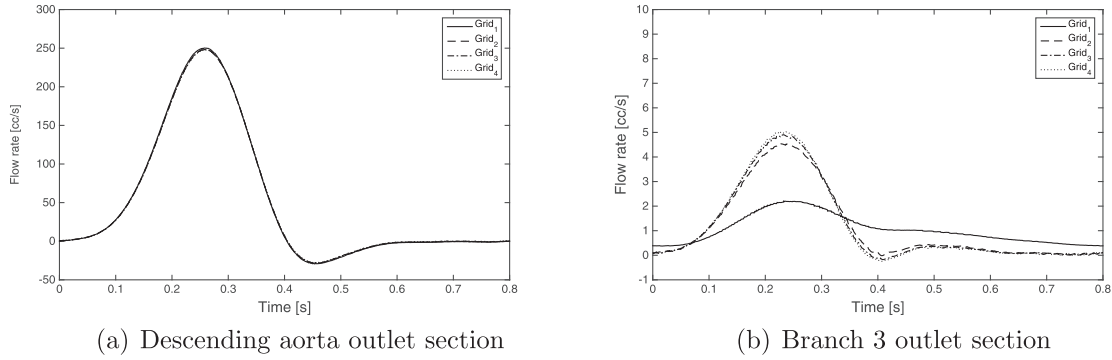


Fig. 30. Flow rates for the four grids used in the grid sensitivity analysis.

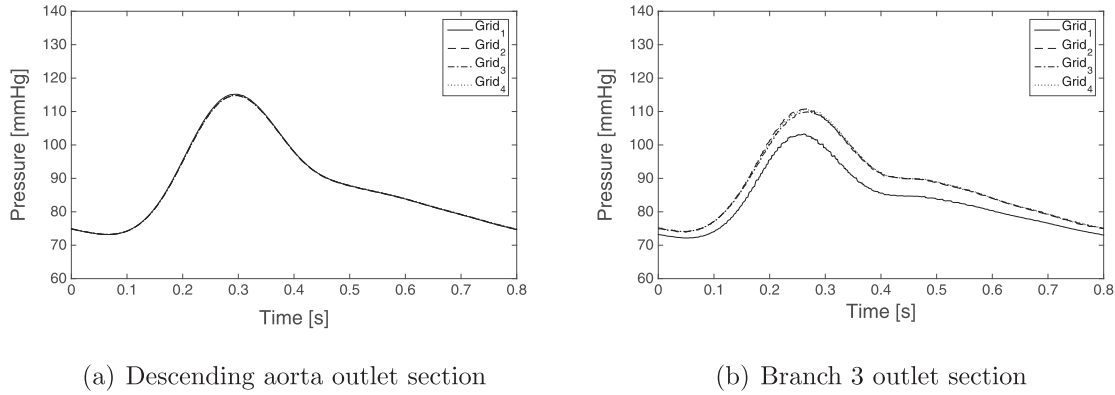


Fig. 31. Average pressures for the four grids used in the grid sensitivity analysis.

outflow parameters, but the same methodology could be applied to different types of uncertainties, as e.g., uncertainties in the geometry due to imaging/reconstruction errors or in the mechanical properties of the vessel walls or in the flow rate waveform at the inlet. These additional sources of uncertainty will be the object of future studies.

Appendix A. Grid sensitivity analysis

As introduced in Section 2.2, we considered four unstructured grids, having approximately 1.1×10^5 , 1.6×10^5 , 2.4×10^5 and 3.9×10^5 nodes. Grid independence was evaluated on the values of several output quantities. First of all, we considered the flow rates and average pressures at all outlet sections. Fig. 30 shows the flow rates at the largest (descending aorta) and smallest (third branch) outlet sections, whereas Fig. 31 considers the average pressures at the same locations. Flow rates and average pressures at the descending aorta outlet section are practically the same for all the four grids considered, see Figs. 30(a) and 31(a). On the other hand, in correspondence of the third branch there are significant differences: in particular, Figs. 30(b) and 31(b) show that Grid₁ is clearly too coarse, and also Grid₂ seems to be not sufficient to capture flow rates (Fig. 30(b)). Grid₃ and Grid₄ give the same results.

The above considerations are confirmed by the L_2 norms of the normalized differences between the results obtained with two subsequent grids, presented in Table 3 for the flow rates and in Table 4 for the average pressures. The normalization was done dividing by the mean value of the considered quantity in the coarser grid.

Then we considered the values of velocity and pressure in the whole domain: in order to compare two subsequent grids, the variable fields obtained on the finer grid were resampled on the

Table 3

Comparison between subsequent grids: L_2 norms of normalized differences of flow rate at outlets.

	Grid ₂ - Grid ₁	Grid ₃ - Grid ₂	Grid ₄ - Grid ₃
Descending aorta	0.0189	0.0082	0.0100
Branch 1	0.0236	0.0246	0.0249
Branch 2	0.0172	0.0231	0.0260
Branch 3	0.9659	0.1287	0.0500

Table 4

Comparison between subsequent grids: L_2 norms of normalized differences of average pressure at outlets.

	Grid ₂ - Grid ₁	Grid ₃ - Grid ₂	Grid ₄ - Grid ₃
Descending aorta	0.0061	0.0025	0.0026
Branch 1	0.0021	0.00215	0.0015
Branch 2	0.0019	0.0024	0.0017
Branch 3	0.0598	0.0053	0.0037

coarser one, and thus the difference was evaluated node by node. Focusing on the comparison between Grid₃ and Grid₄, the largest L_2 norm of the differences over a cardiac cycle was 0.05 m s^{-1} for the velocity field and 1 mmHg for the pressure field: these values were obtained at systolic peak. For sake of comparisons, during systolic peak the values of velocity and average pressure at the inlet section were $\sim 0.5 \text{ m s}^{-1}$ and $\sim 120 \text{ mmHg}$, respectively.

For the previous considerations, we chose Grid₃ for the numerical simulations performed in this work.

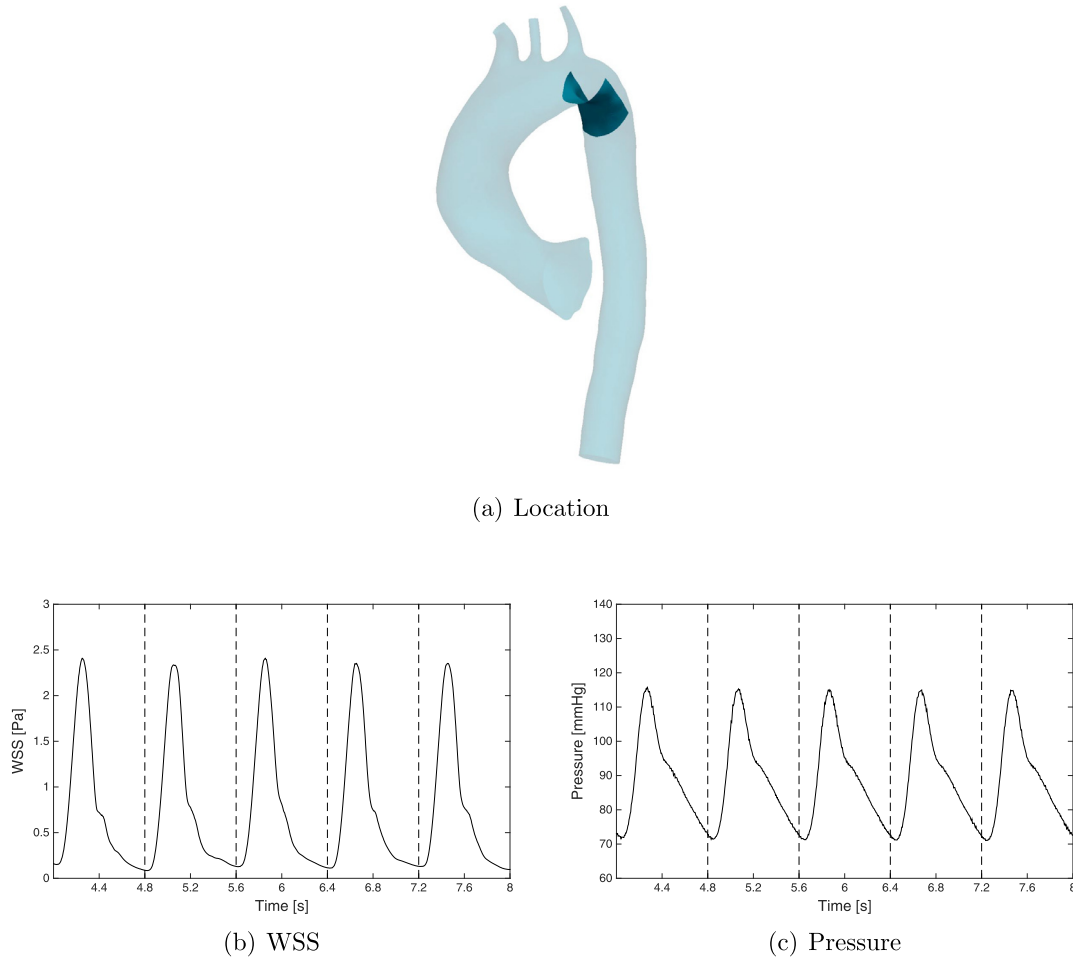


Fig. 32. Location where the signals are recorded (a) and variables behavior during 5 consecutive cycles: WSS (b) and pressure (c) in a representative node in the considered region.

Appendix B. Turbulence analysis

In order to check the possible presence of turbulent fluctuations associated to scales resolvable on the considered grids, we analyzed the periodicity of WSS and pressure during five consecutive cardiac cycles. We chose to evaluate these quantities in some grid nodes located in the internal region of the aortic arch, downstream of the curvature (Fig. 32(a)), because this is a location where flow recirculations typically occur.

Fig. 32(b) and 32(c) show that the two variables are substantially periodic, without noticeable turbulent fluctuations. These figures were obtained in case of Grid₃, i.e., the grid chosen for all the simulations present in this study, but the same behavior was also found using more refined grids.

Appendix C. Convergence of the gPC response surfaces

As in [47], the convergence of the stochastic method was evaluated by considering the coefficients of the gPC expansion (introduced in Section 2.3) and calculating the contribute of the highest-order polynomials to the total variance. The total variance is $V = \sum_{k=1}^T (a_k^*)^2$ with $a_k^* = a_k / |\Phi_k|$, $|\Phi_k|$ being the norm of the polynomial of order k .

Fig. 33 reports the coefficients obtained in the three considered stochastic analyses for the scalar output quantities, i.e., the systolic pressure $P_{\max}$, the diastolic pressure $P_{\min}$, the mean pressure P_{mean} and the pulse pressure ΔP . In all cases the coefficients associated to the highest-order polynomials (i.e., third order) are much lower than those related to lower-order polynomials. Indeed, the contribution of the third-order gPC polynomials to the total variance is always below 1%, except for P_{mean} in the deformable case (represented by empty circle markers in Fig. 33(c)). However, we recall that in this last case the total variability associated to P_{mean} was in fact negligible, as shown in Fig. 22(b).

Similar considerations could also be made for the pressure waveforms and the TAWSS indicators considered in the stochastic analysis: the space-averaged contributions of the third order polynomials remain in all the considered cases lower than 1% and 4% respectively. Considering the instantaneous WSS, the same quantity slightly varied during the cardiac cycle, with a maximum space-averaged contribution of about 6%, observed for the rigid case at the systolic peak instant.

The above results suggest that further increasing the order of the gPC expansion would not produce major contributions to the variance in the output quantities.

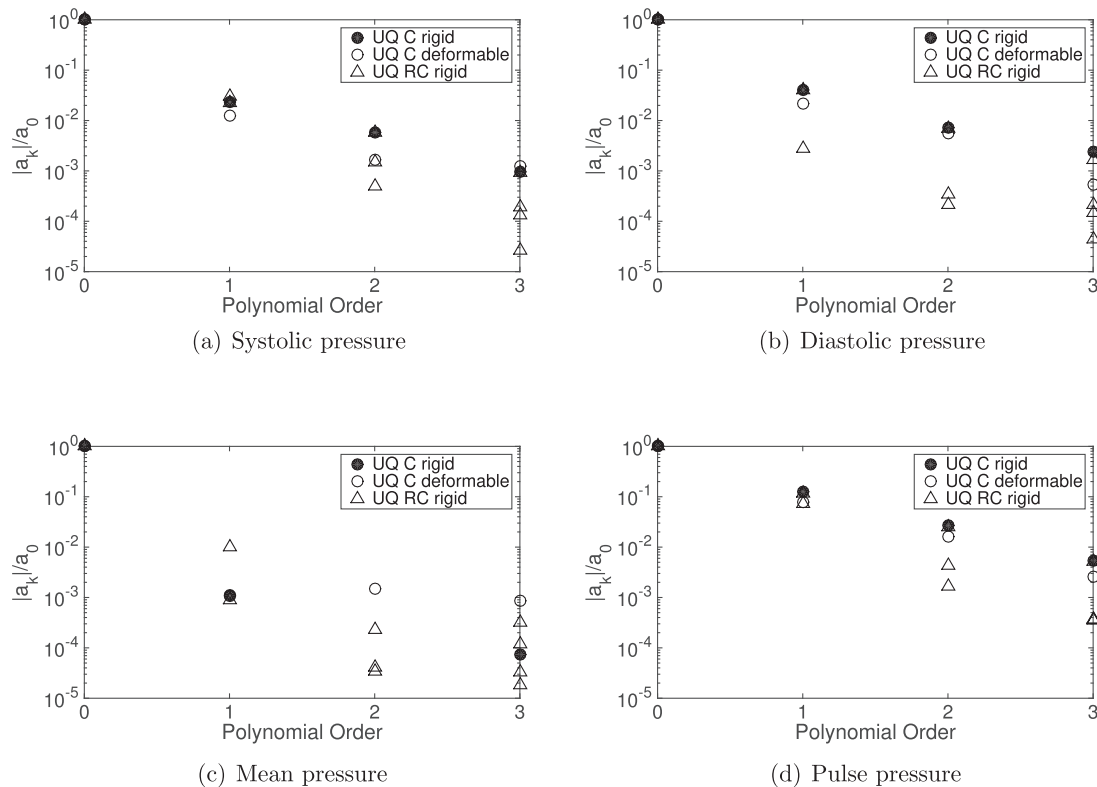


Fig. 33. Coefficients of the gPC polynomial expansion for the performed stochastic analyses.

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